Multiweight arithmetic holonomy and density-one irrationality of p-adic zeta values

Christopher D. Long

All-prime dependency-audited exact-capacity revision — 26 August 2026

Abstract

For each genus-zero prime

$$\mathcal{P}_0 = \{2, 3, 5, 7, 13\}$$

and each finite set $\mathcal{S}$ of distinct odd integers at least 3, we combine the Eisenstein–Eichler extensions attached to the values $\zeta_p(s)$, $s \in \mathcal{S}$, using one common homogenizing coordinate. Under the joint rationality hypothesis the resulting rank-m horizontal leaf, where $m = 1 + \sum_{s\in\mathcal{S}} s$, has full functional rank. The blockwise genuine-source Hasse kernels and a common torsor-valued Cartier argument give the Hasse-improved multiweight type

$$\tau^{\mathrm{H}}_{p,\mathcal{S}}(\xi) = \frac{2}{m^2}\left[\xi \sum_{s\in\mathcal{S}} s^2 - \left(\sum_{s\in\mathcal{S}}(s-1)\right) J_p(\xi) + (\max \mathcal{S}) I^m_\xi(\xi)\right].$$

A thickness-sensitive Fitting refinement replaces the rectangular pole bound by a denominator polygon. The certified finite conclusions are unchanged: among the first 1200 positive odd arguments, at least

$$1194, \quad 1191, \quad 1185, \quad 1180, \quad 1061$$

of the values $\zeta_p(s)$ are irrational for $p = 2, 3, 5, 7, 13$, respectively.

We then develop an intrinsic-curve version that no longer assumes $X_0(p) \simeq \mathbf{P}^1$. On the fixed coarse modular curve, after removing the finite elliptic locus, a finite-map restriction-of-scalars argument gives a Shidlovsky zero estimate and exact $q \mapsto q^\ell$ divided Frobenius gives a general-curve q-jet Cartier window. A normalization-robust slopes argument applies finite-order capacitary estimates only after contraction to a scalar section. It yields

$$\tau^{\mathrm{curve}}_{p,\mathcal{S}}(\xi) \geq \Lambda^\circ_p - \frac{C_p}{m},$$

where $\Lambda^\circ_p > 0$ comes from the strict capacitary gain of a p-adic annular collar and C_p is independent of the packet.

We also construct the canonical exact metric. Its finite components are the Bost–Chambert-Loir capacitary $\mathbf{Q}$-model metrics, while its Archimedean components are the Bost–Charles direct-image Green functions of the chosen continuation maps. After clearing the finitely many rational exponents, the capacitary tangent line is the normal line of a pseudoconcave formal-analytic surface. The Chen–Moriwaki determinant Hilbert–Samuel theorem and a uniform finite-order Green–Jensen estimate then give the exact source formula and the target-normal cancellation

$$\widehat{\deg} M_D = \frac{m}{2}\mathscr{H}_\alpha D^2 + o(D^2), \qquad \tau^{\mathrm{curve}}_{p,\mathcal{S}}(\xi) \geq \Lambda_\alpha - \frac{\mathscr{H}_\alpha}{m}.$$

The stronger exact theorem is not needed for the normalization-robust qualitative proof. We then compute the maximal modular capacity pair exactly at every prime. For $0 < Y < 1/p$,

$$(\Lambda_p(Y), \mathscr{H}_p(Y)) = \left(\frac{12\log p}{p-1} - 2\pi Y, \frac{12\log p}{p-1} - 2\pi Y + C_p(Y)\right).$$

For $p \geq 5$ the finite contribution is the effective resistance of the supersingular reduction graph, evaluated by the Eichler–Deuring mass formula; $p = 2, 3$ are treated directly with their integral Hauptmodul lemniscates. If d_p is the degree of the finite map used in the general-curve zero estimate, then

$$R_p(X) := \#\{3 \leq s \leq X : \ s \text{ odd and } \zeta_p(s) \in \mathbf{Q}\} \leq \left(\frac{e\, d_p(p-1)}{12 \log p} + o_p(1) \right) (\log X)^2.$$

Thus the irrational values have natural density one with an explicit leading constant at every fixed prime. A fixed-point finite-packet certificate gives, among the first 1200 odd arguments, at least

$$1010, 974, 963, 823, 786, 775, 743, 605, 594$$

irrational values for $p = 11, 17, 19, 23, 29, 31, 37, 41, 43$, respectively.

Audit status. The genus-zero part isolates the same published interfaces as the underlying single-weight arithmetic-holonomy construction: Calegari's Eisenstein family, Buzzard continuation, Katz's Hasse invariant, the de Rham frame torsor of Graham–Pilloni–Rodrigues Jacinto, and the Bost–Charles/CDT slopes and prime-window estimates. The all-prime extension uses Bost–Chambert-Loir capacitary metrics and four internal bridges: coarse-moduli descent and finite-map Shidlovsky restriction of scalars, intrinsic q-jet Cartier strictness, scalar finite-order capacitary jet bounds with a fixed-twist source comparison, and strict positivity of the distinguished p-adic capacitary gain. The compatibility of these bounds with the hybrid finite-prime lattices is proved explicitly.

The exact normalization is proved separately in Theorem 13.7 and Section A. The audit distinguishes two assertions that had previously been conflated. Bost–Chambert-Loir provides one adelic Arakelov $\mathbf{Q}$-divisor inducing all finite capacitary metrics; after denominator clearing, its metrized tangent line is realizable as the normal line of a pseudoconcave Bost–Charles formal-analytic surface. We do not assert that the entire finite vertical divisor is literally a Bost–Charles direct image, nor that matching the normal line automatically produces a Bost–Charles meromorphic map to the modular curve. Instead, the exact source and jet formulas are proved directly on the modular curve for the canonical mixed capacitary/direct-image divisor, using the Chen–Moriwaki determinant Hilbert–Samuel theorem and a place-by-place finite-order Green–Jensen estimate. The exact modular capacity pair is computed for every prime in Theorem 13.11. Its maximal-domain passage uses the rational skeleton exhaustion of Lemma 13.10; Bost–Chambert-Loir Proposition 5.6 is used only to realize each fixed inner domain as a rational lemniscate, not as a continuity theorem. Supersingular residue fields are split by normalized unramified base change. Consequently the all-prime density coefficient is explicit in terms of the finite-map degree d_p. The positive-genus finite-packet claims use the certified upper bound $2g_p \geq d_p$ and are independently fixed-point certified. Independent specialist review remains appropriate before publication.

Contents

1 Introduction and main results

For an odd integer $s \geq 3$, let $\zeta_p(s)$ denote the Kubota–Leopoldt p-adic zeta value in Calegari's normalization. The single-weight Eisenstein–Eichler method starts from

$$K_{p,s}(q) = \frac{\zeta_p(s)}{2} + \sum_{n \geq 1} \sigma_{-s}^{(p)}(n) q^n, \qquad \sigma_{-s}^{(p)}(n) = \sum_{\substack{d \mid n \\ p \nmid d}} d^{-s}.$$

Under the contrary hypothesis $\zeta_p(s) \in \mathbf{Q}$, this is an arithmetic horizontal leaf of a rank-$(s+1)$ algebraic connection. The present paper uses several such blocks simultaneously but shares their scalar coordinate. The resulting rank is essentially the sum of the weights, while the largest large-prime pole grade is only their maximum.

Let $\mathcal{S}$ be a finite nonempty set of distinct odd integers $s \geq 3$. Put

$$m = 1 + \sum_{s \in \mathcal{S}} s, \qquad Q_{\mathcal{S}} = \sum_{s \in \mathcal{S}} s^2, \qquad B_{\mathcal{S}} = \sum_{s \in \mathcal{S}} (s-1), \qquad s_* = \max \mathcal{S}. \tag{1}$$

For $p \in \{2, 3, 5, 7, 13\}$, define

$$c_p = \frac{p+1}{12}, \qquad J_p(\xi) = \int_0^{\xi} (1 - c_p x)_+ \, \mathrm{d}x. \tag{2}$$

Thus

$$J_p(\xi) = \begin{cases} \xi - \dfrac{p+1}{24}\xi^2, & c_p\xi \le 1, \\ \dfrac{6}{p+1}, & c_p\xi \ge 1. \end{cases}$$

In particular, because $c_{13} = 7/6$, every admissible cutoff $\xi > 1$ satisfies

$$J_{13}(\xi) = \frac{3}{7}. \tag{3}$$

For $m \ge 2$ and $1 < \xi < m$, put

$$K_m(\xi) = \left\lfloor \frac{m-1}{\xi} \right\rfloor, \tag{4}$$

and

$$I_\xi^m(\xi) = \left(m - \frac{1}{2}\right) H_{K_m(\xi)} - K_m(\xi)\xi + \frac{(m - (K_m(\xi)+1)\xi)_+^2}{2(K_m(\xi)+1)}. \tag{5}$$

Theorem 1.1 (Hasse-improved multiweight arithmetic holonomy)**.** *Fix $p \in \{2, 3, 5, 7, 13\}$ and a finite set $\mathcal{S}$ as above. Assume*

$$\zeta_p(s) \in \mathbf{Q} \qquad (s \in \mathcal{S}).$$

Let $\boldsymbol{\phi}$ be admissible adelic continuation templates for the simultaneous Eisenstein–Eichler leaf, with derivative width $\Lambda(\boldsymbol{\phi})$ and Bost–Charles energy $\mathscr{H}(\boldsymbol{\phi})$. For every $1 < \xi < m$,

$$m\left(\Lambda(\boldsymbol{\phi}) - \tau_{p,\mathcal{S}}^{\mathrm{H}}(\xi)\right) \le \mathscr{H}(\boldsymbol{\phi}), \tag{6}$$

where

$$\boxed{\tau_{p,\mathcal{S}}^{\mathrm{H}}(\xi) = \frac{2}{m^2}\left[\xi Q_{\mathcal{S}} - B_{\mathcal{S}} J_p(\xi) + s_* I_\xi^m(\xi)\right].} \tag{7}$$

The large-prime term in (7) is deliberately rectangular: it charges every vulnerable pivot at the largest possible pole grade s_*. A stronger theorem records the finite ℓ-adic thickness available above each pole grade.

For $1 \le b \le s_*$ put

$$\rho_b = \sum_{\substack{s \in \mathcal{S} \\ s \ge b}} s(s - b + 1). \tag{8}$$

Let

$$w_m(x) = K_x + (m - (K_x+1)x)_+, \qquad K_x = \left\lfloor \frac{m-1}{x} \right\rfloor,$$

and define the capped window

$$I_\xi^{m,\rho}(\xi) = \int_\xi^m \min\{w_m(x), \rho\} \, \mathrm{d}x. \tag{9}$$

Theorem 1.2 (Thickness-sensitive denominator polygon)**.** *Under the hypotheses of Theorem 1.1, joint rationality also forces*

$$m\left(\Lambda(\phi) - \tau^{\mathrm{poly}}_{p,\mathcal{S}}(\xi)\right) \le \mathscr{H}(\phi), \tag{10}$$

where

$$\boxed{\tau^{\mathrm{poly}}_{p,\mathcal{S}}(\xi) = \frac{2}{m^2}\left[\xi Q_{\mathcal{S}} - B_{\mathcal{S}} J_p(\xi) + \sum_{b=1}^{s_*} I^{m,\rho_b}_{\xi}(\xi)\right].} \tag{11}$$

Moreover

$$I^{m,\rho}_{\xi}(\xi) = \begin{cases} \left(m - \dfrac{1}{2}\right) H_\rho - \rho\xi, & 1 < \xi < \dfrac{m}{\rho+1}, \\ I^m_\xi(\xi), & \xi \ge \dfrac{m}{\rho+1}. \end{cases} \tag{12}$$

In particular, $\tau^{\mathrm{poly}}_{p,\mathcal{S}}(\xi) \le \tau^{\mathrm{H}}_{p,\mathcal{S}}(\xi)$.

The counting statements are certified from Theorem 1.1, not from the refinement.

Theorem 1.3 (Explicit simultaneous irrationality bounds)**.** *Among the* 1200 *values*

$$\zeta_p(3), \zeta_p(5), \ldots, \zeta_p(2401),$$

the following numbers are irrational:

p	*max. rational*	*min. irrational*	*cutoff for* 1000
2	6	1194	$s \le 2013$
3	9	1191	$s \le 2019$
5	15	1185	$s \le 2029$
7	20	1180	$s \le 2039$
13	139	1061	$s \le 2279$

The cutoff in the last column means that at least 1000 *of the positive odd values with argument at most that cutoff are irrational. The first four rows are certified by the branch-and-bound program; the* $p = 13$ *row follows from the elementary uniform estimate in Proposition 9.2.*

Theorem 1.4 (Certified positive-genus finite-packet bounds)**.** *For the positive-genus prime levels* $p \le 43$*, the following bounds hold among*

$$\zeta_p(3), \zeta_p(5), \ldots, \zeta_p(2401).$$

The quantity $\bar{d}$ *is the safe upper bound* $\bar{d} = 2g_p$ *for the finite-map degree* d_p *supplied by Theorem 11.2.*

p	g_p	$\bar{d}$	*max. rational*	*min. irrational*	*cutoff for* 1000
11	1	2	190	1010	$s \le 2379$
17	1	2	226	974	$s \le 2459$
19	1	2	237	963	$s \le 2483$
23	2	4	377	823	$s \le 2823$
29	2	4	414	786	$s \le 2921$
31	2	4	425	775	$s \le 2951$
37	2	4	457	743	$s \le 3039$
41	3	6	595	605	$s \le 3445$
43	3	6	606	594	$s \le 3483$

The last column means that the prefix ending at the displayed odd argument contains at least 1000 *irrational values. These bounds use only the exact all-prime capacity pair, the general-curve arithmetic type, and the elementary packet majorants of Proposition 14.2; they do not use an unproved low-degree map or a genus-zero Hasse saving.*

For every prime p, put

$$R_p(X) = \#\{3 \le s \le X : \ s \text{ odd and } \zeta_p(s) \in \mathbf{Q}\}. \tag{13}$$

Theorem 1.5 (Quantitative genus-zero density one)**.** *For* $p \in \{2, 3, 5, 7, 13\}$*, one has*

$$\boxed{R_p(X) \le \left(\frac{e(p-1)}{12 \log p} + o(1)\right) (\log X)^2.} \tag{14}$$

Consequently

$$\#\{3 \le s \le X : \ s \text{ odd and } \zeta_p(s) \notin \mathbf{Q}\} = \frac{X}{2} - O_p((\log X)^2),$$

and the irrational values have natural density one among the positive odd arguments.

Theorem 1.6 (Explicit density-one irrationality at every fixed prime)**.** *Let* d_p *be the degree of the finite map in Theorem 11.2; one may take* $d_p = 1$ *when* $g_p = 0$ *and* $d_p \le 2g_p$ *when* $g_p \ge 1$*. For every fixed prime* p*,*

$$\boxed{R_p(X) \le \left(\frac{e\, d_p(p-1)}{12 \log p} + o_p(1)\right) (\log X)^2.} \tag{15}$$

In particular, when $g_p \ge 1$*,*

$$R_p(X) \le \left(\frac{e\, g_p(p-1)}{6 \log p} + o_p(1)\right) (\log X)^2. \tag{16}$$

Consequently

$$\#\{3 \le s \le X : \ s \text{ odd and } \zeta_p(s) \notin \mathbf{Q}\} = \frac{X}{2} - O_p((\log X)^2),$$

and the irrational values have natural density one among the positive odd arguments. At genus zero, $d_p = 1$ *and the leading coefficient agrees with* (14)*.*

The induction begins with the following individual theorem, which is also a special case of the single-weight machinery developed below.

Theorem 1.7 (Base individual values)**.** *The following values are irrational:*

$$\zeta_2(s) \quad (s = 3, 5, \ldots, 29), \qquad \zeta_3(s) \quad (s = 3, 5, \ldots, 11),$$

$$\zeta_5(3), \quad \zeta_5(5), \quad \zeta_7(3).$$

Warning 1.8 (Logical scope). All asymptotics are for a fixed finite packet $\mathcal{S}$ while $D \to \infty$. The constants and exceptional primes may depend on p and $\mathcal{S}$. This is sufficient for every counting contradiction: after assuming that a specified number of rational values exist, one fixes one such finite set and then runs the asymptotic argument.

2 Published inputs and normalization

We use the following external results.

(E1) **Overconvergent Eisenstein families.** For $s = 2n+1$, Calegari's specialization at the character $\kappa(a) = a^{1-s}$ gives a weight-$(1-s)$ overconvergent eigenform whose nonconstant coefficients are $\sigma_{-s}^{(p)}(n)$ and whose constant term is $\zeta_p(s)/2$; see [6, Theorem 2.3 and Lemma 2.4].

(E2) **Finite-slope continuation.** Buzzard's theorem extends a finite-slope overconvergent form across the wide-open $W_0(p)$; see [5, Theorem 5.2]. The relevant radii for $p = 2, 3, 5, 7, 13$ are recalled in Section 8.

(E3) **Hasse invariants.** We use Katz's geometric Hasse invariant and q-expansion principle [11], together with

$$E_{\ell-1} \equiv 1 \pmod{\ell}, \qquad E_{\ell+1} \equiv E_2 \pmod{\ell}.$$

(E4) **De Rham frame torsor and Gauss–Manin connection.** We use the Hodge-compatible B-torsor and connection of [10, Propositions A.14–A.15]. The descent to one fixed coarse modular curve and the uniform spreading-out statements used below are proved internally.

(E5) **Arithmetic holonomy.** We use Bost's slopes inequality [2]; the capacitary metrics and model-theoretic Dirichlet problem of [3]; the theta Hilbert–Samuel, overflow, and direct-image divisor formulas of [4]; the determinant vector-bundle Hilbert–Samuel theorem of [9, Corollary 4.5.2]; and the zero-estimate, source-degree, evaluation-height, and prime-window formulas in [7, 8]. Regularity of the modular stack at the cusps is taken from [12]. The genus-zero fixed-bundle bridge and all intrinsic-curve passages needed at arbitrary prime level are proved internally in Sections 7 and 11 to 13.

Put

$$f_p(\tau) = \left(\frac{\Delta(p\tau)}{\Delta(\tau)}\right)^{1/(p-1)}. \tag{17}$$

For $p \in \{2, 3, 5, 7, 13\}$ this is a Hauptmodul at the cusp $P = \infty$. For $p = 13$, Maier's normalization [15, (3.2) and Table 4] gives the canonical Hauptmodul $t_{13} = 13[13]^2/[1]^2 = 13 f_{13}$ and its degree-14 map to the j-line. In all five cases

$$f_p(q) = q \prod_{n \geq 1} \left(1 + q^n + \cdots + q^{(p-1)n}\right)^{24/(p-1)} \in q + q^2\mathbf{Z}[[q]]. \tag{18}$$

Its inverse satisfies

$$q(f_p) \in f_p + f_p^2\mathbf{Z}[[f_p]]. \tag{19}$$

Throughout the genus-zero sections we write $f = f_p$. In the arbitrary-prime sections no Hauptmodul is used; all local arithmetic is performed in the width-one Tate parameter q. In either notation,

$$D = q\frac{\mathrm{d}}{\mathrm{d}q}.$$

For admissible adelic continuation templates $\boldsymbol{\phi} = (\phi_v)_v$, centered at P, define

$$\Lambda(\boldsymbol{\phi}) = \sum_v \log |\phi_v'(0)|_v \tag{20}$$

and

$$\mathscr{H}(\boldsymbol{\phi}) = \sum_{v|\infty} \iint_{\mathbb{T}_v^2} \log |\phi_v(z) - \phi_v(w)|_v \, \mathrm{d}m(z)\,\mathrm{d}m(w) + \sum_{v\nmid\infty} \sup_{z\in\mathbb{T}_v} \log |\phi_v(z)|_v \,. \tag{21}$$

These are normalized as in [7, Section 7.3] and [8, Section 2].

3 Single-weight Eisenstein–Eichler blocks

Fix an odd $s \geq 3$. Define

$$J_{p,s}(q) = \sum_{n \geq 1} \sigma_{-s}^{(p)}(n) q^n, \qquad \sigma_{-s}^{(p)}(n) = \sum_{\substack{d|n \\ p\nmid d}} d^{-s}, \tag{22}$$

and, for a scalar η,

$$K_{p,s,\eta} = J_{p,s} + \eta.$$

At $\eta = \zeta_p(s)/2$, input (E1) identifies this with the distinguished overconvergent form of weight $1 - s$.

Lemma 3.1 (U_p eigenvalue)**.** *For every $n \geq 1$,*

$$\sigma_{-s}^{(p)}(pn) = \sigma_{-s}^{(p)}(n).$$

Consequently

$$U_p K_{p,s,\zeta_p(s)/2} = K_{p,s,\zeta_p(s)/2}.$$

Proof. The divisors of pn prime to p are exactly the divisors of n prime to p. The usual q-expansion definition of U_p gives the assertion, including the constant term. □

For $1 \leq e \leq s$ put

$$R_{s,e} = D^{s-e} J_{p,s}. \tag{23}$$

When no confusion is possible we write R_e.

Lemma 3.2 (Raw integration depth)**.** *For every $n \geq 1$,*

$$n^e [q^n] R_{s,e} = \sum_{\substack{d|n \\ p\nmid d}} (n/d)^s \in \mathbf{Z}. \tag{24}$$

If $L_N = \operatorname{lcm}(1, \ldots, N)$, then

$$L_N^e R_{s,e}(q(f)) \pmod{f^{N+1}} \in \mathbf{Z}[[f]]/(f^{N+1}).$$

Proof. The coefficient of q^n is $n^{s-e}\sigma_{-s}^{(p)}(n)$, which gives (24). Since $n \mid L_N$ for $n \leq N$, the q-integrality follows. The integral tangent-to-identity substitution (19) uses only coefficients of index at most the target index and preserves the cumulative cutoff. □

Lemma 3.3 (Exact divided Frobenius)**.** *Let $\ell \neq p$ be prime. For $1 \leq e \leq s$,*

$$R_{s,e}(q) - \ell^{-e} R_{s,e}(q^\ell) \in \mathbf{Z}_\ell[[q]]. \tag{25}$$

After $q = q(f)$,

$$R_{s,e}(q(f)) - \ell^{-e} R_{s,e}(q(\varphi_\ell(f))) \in \mathbf{Z}_\ell[[f]], \tag{26}$$

where

$$\varphi_\ell(f) = f(q(f)^\ell).$$

Moreover, outside a fixed exceptional set,

$$\varphi_\ell(f) = f^\ell + \ell \Delta_\ell(f), \qquad \Delta_\ell(f) \in f^{\ell+1} \mathbf{Z}_\ell[[f]]. \tag{27}$$

Proof. If $\ell \mid n$, the q^n coefficient of the left side of (25) is

$$n^{s-e}\left(\sigma^{(p)}_{-s}(n) - \ell^{-s}\sigma^{(p)}_{-s}(n/\ell)\right) = n^{s-e}\sum_{\substack{d\mid n\\ p\nmid d\\ \ell\nmid d}} d^{-s},$$

which is ℓ-integral; the case $\ell \nmid n$ is immediate. Substitution gives (26). Finally, in characteristic ℓ, $f(q)^\ell = f(q^\ell)$. Substituting $q = q(f)$ gives $\varphi_\ell(f) \equiv f^\ell \pmod{\ell}$; the two sides have the same order and leading coefficient, proving (27). □

Let $\mathcal{H}$ be the rank-two de Rham bundle of the universal elliptic curve and put

$$V_s = \operatorname{Sym}^{s-1}\mathcal{H}.$$

In a local oper frame $v_0, \ldots, v_{s-1}$ with

$$\nabla_\partial v_j = (s-1-j)v_{j+1}, \qquad v_s = 0,$$

define

$$\operatorname{spl}_s(F) = \sum_{j=0}^{s-1}(-1)^j \frac{(s-1)!}{(s-1-j)!}\partial^{s-1-j}F\, v_j. \tag{28}$$

Lemma 3.4 (Canonical BGG lift)**.** *The expression* (28) *glues to an algebraic differential operator*

$$\operatorname{spl}_s : \omega^{1-s} \longrightarrow V_s$$

and satisfies

$$\nabla_{V_s}\operatorname{spl}_s(F) = \iota_s(\operatorname{Bol}_s F). \tag{29}$$

On the Tate disc, $\operatorname{Bol}_s F = D^s F$*.*

Proof. Write $c_j = (-1)^j(s-1)!/(s-1-j)!$. In $\nabla_\partial \operatorname{spl}_s(F)$, the coefficient of $\partial^{s-j}F v_j$ for $j \geq 1$ is $c_j + (s-j)c_{j-1} = 0$. Only the top-oper term remains. The formula is characterized by this property and its bottom quotient, hence glues. The Tate normalization is the standard dual-BGG/Bol normalization; compare [16, Proposition 2.4.3] and [17, Proposition 2.4.1]. □

Put

$$E^{\mathrm{evil}}_{p,s+1} = D^s J_{p,s}. \tag{30}$$

It is the algebraic p-stabilized Eisenstein form of weight $s+1$ and is independent of η.

Proposition 3.5 (Split Eisenstein–Eichler connection)**.** *On*

$$\mathcal{E}_{p,s} = \mathcal{O} \oplus V_s$$

define

$$\nabla_{\mathcal{E}}(a, v) = \left(\mathrm{d}a, \nabla_{V_s}v - a\,\iota_s(E^{\mathrm{evil}}_{p,s+1})\right). \tag{31}$$

This is an algebraic regular-singular connection, and

$$S_{p,s,\eta} = (1, \operatorname{spl}_s(K_{p,s,\eta})) \tag{32}$$

is horizontal on the Tate disc.

Proof. The connection is algebraic because $E^{\mathrm{evil}}_{p,s+1}$ is algebraic; integrability is automatic on a curve. Equation (29) gives horizontality. □

Proposition 3.6 (Single-weight functional rank)**.** *The complex monodromy orbit of $S_{p,s,\eta}$ spans a space of dimension $s+1$.*

Proof. The period-polynomial cocycle is nonzero. Otherwise one could subtract a polynomial of degree at most $s-1$ from the Eichler primitive and obtain a moderate-growth modular form of weight $1-s<0$, hence zero, contradicting $D^sK = E^{\mathrm{evil}}_{p,s+1} \neq 0$. The span of the cocycle values is $\Gamma_0(p)$-stable. Since $\Gamma_0(p)$ is Zariski dense in SL_2 and $\mathrm{Sym}^{s-1}(\mathbf{C}^2)$ is irreducible, that span is the full s-dimensional representation. Adjoining the scalar coordinate gives rank $s+1$. ☐

4 The simultaneous packet and its zero estimate

Assume from now on that $\mathcal{S}$ is fixed and that

$$\eta_s = \frac{\zeta_p(s)}{2} \in \mathbf{Q} \qquad (s \in \mathcal{S}). \tag{33}$$

Define

$$\mathcal{E}_{p,\mathcal{S}} = \mathcal{O} \oplus \bigoplus_{s\in\mathcal{S}} V_s \tag{34}$$

with

$$\nabla_{\mathcal{S}}(a,(v_s)_s) = \Big(\mathrm{d}a, \Big(\nabla_{V_s} v_s - a\,\iota_s(E^{\mathrm{evil}}_{p,s+1})\Big)_s\Big). \tag{35}$$

Then

$$S_{p,\mathcal{S}} = (1, (\mathrm{spl}_s(K_{p,s,\eta_s}))_{s\in\mathcal{S}}) \tag{36}$$

is horizontal.

Theorem 4.1 (Joint functional rank)**.** *The complex monodromy orbit of $S_{p,\mathcal{S}}$ spans the full fibre of $\mathcal{E}_{p,\mathcal{S}}$. Its dimension is*

$$m = 1 + \sum_{s\in\mathcal{S}} s.$$

Proof. Fix a basepoint and put

$$V_{\mathcal{S}} = \bigoplus_{s\in\mathcal{S}} V_s.$$

Let $W \subset V_{\mathcal{S}}$ be the span of all differences

$$\rho_{\mathcal{S}}(\gamma)S_{p,\mathcal{S}} - S_{p,\mathcal{S}}.$$

The scalar coordinate of each difference is zero. For $\alpha, \gamma \in \Gamma_0(p)$,

$$\rho_V(\alpha)\left(\rho_{\mathcal{S}}(\gamma)S - S\right) = \left(\rho_{\mathcal{S}}(\alpha\gamma)S - S\right) - \left(\rho_{\mathcal{S}}(\alpha)S - S\right),$$

so W is $\Gamma_0(p)$-stable. Projection to the s-block is the span of the orbit differences of the single-weight leaf; by Proposition 3.6, it is all of V_s.

The stabilizer of a linear subspace is Zariski closed, so density of $\Gamma_0(p)$ makes W an SL_2-subrepresentation. The representations $V_s = \mathrm{Sym}^{s-1}(\mathbf{C}^2)$ are pairwise nonisomorphic irreducibles because the weights in $\mathcal{S}$ are distinct. Their direct sum is multiplicity-free. A subrepresentation surjecting onto every summand must be the entire direct sum. Hence $W = V_{\mathcal{S}}$. One branch of S has scalar coordinate 1, so it is independent of W and completes the span. ☐

Corollary 4.2 (Joint functional independence)**.** *In any rational algebraic frame, the m coordinate functions of $S_{p,\mathcal{S}}$ are linearly independent over the function field $\mathbf{C}(X_0(p))$. For $p \in \mathcal{P}_0$ this is $\mathbf{C}(f_p)$, and hence the coordinates are also independent over $\mathbf{Q}(f_p)$.*

Proof. A rational covector over $\mathbf{C}(X_0(p))$ annihilating one branch is single-valued, hence annihilates all analytic continuations. At a generic basepoint those continuations span the full fibre by Theorem 4.1; the covector is zero. $\square$

Choose a fixed divisor B, disjoint from P, clearing all fixed poles, and put

$$M_D = H^0\left(X_0(p), \mathcal{E}_{p,\mathcal{S}}^{\vee}(DP + B)\right). \tag{37}$$

Corollary 4.3 (Genus-zero joint Bost pivot set)**.** *For every $\varepsilon > 0$,*

$$\# V_D = mD + O_{p,\mathcal{S}}(1), \qquad V_D \subset [0, (m+\varepsilon)D + O_{p,\mathcal{S},\varepsilon}(1)]. \tag{38}$$

Proof. Since $X_0(p) \simeq \mathbf{P}^1$ for the five primes in $\mathcal{P}_0$, Riemann–Roch gives $\dim M_D = mD + O(1)$. A fixed rational frame and fixed normalizer embed M_D into

$$(\mathbf{Q}[f]_{\leq D+C})^m.$$

The coordinate functions are holonomic and independent over $\mathbf{C}(f)$ by Corollary 4.2. The Chudnovsky–Osgood estimate in [7, Theorem 3.2.13] bounds the order of a nonzero evaluation by $(m+\varepsilon)D + O(1)$. Evaluation is injective, and every nonzero graded piece of the one-variable vanishing filtration has dimension one, so the number of jumps equals the source rank. $\square$

5 Global sources and the blockwise Hasse saving

The split bundle yields

$$M_D = M_D^0 \oplus \bigoplus_{s \in \mathcal{S}} M_{D,s}^+, \tag{39}$$

where

$$M_D^0 = H^0(X_0(p), \mathcal{O}(DP + B)), \qquad M_{D,s}^+ = H^0(X_0(p), V_s^{\vee}(DP + B)).$$

Riemann–Roch gives

$$\operatorname{rank} M_{D,s}^+ = sD + O_{p,s}(1). \tag{40}$$

After enlarging one fixed exceptional set, this decomposition and the relevant filtrations are saturated over every good $\mathbf{Z}_\ell$.

We recall the blockwise Hasse mechanism in a form that can be summed over $\mathcal{S}$. Write $s = 2r + 1$ and

$$H = D \log f.$$

On a fixed algebraic Hodge splitting, let X be Urban's unipotent coordinate and put $x = X/H$.

Proposition 5.1 (Single unipotent action)**.** *For every $D \gg 1$ and every $\lambda \in M_{D,s}^+$, the multiplier of the normalized undifferentiated row $H^r K_{p,s,\eta_s}$ in the genuine scalar contraction has the form*

$$q_{0,s,\lambda}(f, x) = \sum_{j=0}^{s-1} Q_{j,s,\lambda}(f) x^j. \tag{41}$$

There are a fixed polynomial $g_s(f)$, with $g_s(0)$ a unit, and a constant C_s such that

$$g_s Q_{j,s,\lambda} \in \mathbf{Z}[\Sigma^{-1}][f], \qquad \deg(g_s Q_{j,s,\lambda}) \leq D + C_s. \tag{42}$$

Only one degree-$(s-1)$ unipotent matrix occurs.

Proof. In the monomial basis of $\mathrm{Sym}^{s-1}\mathcal{H}$, the matrix of $\left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right)$ and its dual have entries in $\mathbf{Z}[x]$ of degree at most $s-1$. Keep the source covector in the fixed algebraic dual frame and transport the BGG section once. Equivariance of contraction moves that one unipotent matrix to either factor; it does not create a second action. The undifferentiated K occurs only in the bottom extremal BGG component. Multiplication by f^D clears the variable pole at P, while one fixed polynomial clears all remaining poles and the fixed normalization by H^{-r}. $\square$

On the chosen splitting,

$$x_\sigma = -\frac{E_2}{12H} + r_\sigma(f), \qquad r_\sigma(f) \in \mathbf{Q}(f). \tag{43}$$

Lemma 5.2 (Hasse algebraization)**.** *For every sufficiently good auxiliary prime ℓ, the reduction of x_σ lies in $\mathbf{F}_\ell(f)$. If*

$$\bar{x}_\sigma = \frac{a_\ell(f)}{b_\ell(f)}$$

is in lowest terms and $b_\ell(0) \neq 0$, then

$$\delta_\ell := \max\{\deg a_\ell, \deg b_\ell\} \leq c_p\ell + C_s. \tag{44}$$

Proof. Modulo ℓ, represent E_2 by $E_{\ell+1}/E_{\ell-1}$. Equation (43) becomes a weight-zero modular function. The valence formula on $X_0(p)$ bounds the zero and pole degree by $[\mathrm{SL}_2(\mathbf{Z}) : \Gamma_0(p)](\ell+1)/12 = (p+1)(\ell+1)/12$; the fixed rational correction and elliptic-point conventions change only $O_{p,s}(1)$. Regularity at the Tate cusp gives $b_\ell(0) \neq 0$. $\square$

Theorem 5.3 (Blockwise genuine-source Hasse kernel)**.** *There are constants D_0, C_s, independent of D and of every good ℓ, such that the reduction of $M^+_{D,s}$ contains a subspace $\overline{K}_{D,\ell,s}$ of dimension at least*

$$(s-1)\max\{0, D - \lceil c_p\ell\rceil - C_s\} \tag{45}$$

for which

$$q_{0,s,\bar{\lambda}}\left(f, \frac{a_\ell}{b_\ell}\right) = 0 \quad \textit{in } \mathbf{F}_\ell[[f]]. \tag{46}$$

It lifts to a saturated direct summand $K_{D,\ell,s} \subset M^+_{D,s,\ell}$ with

$$q_{0,s,\lambda}(f, x_\sigma(f)) \in \ell\mathbf{Z}_\ell[[f]]. \tag{47}$$

The construction is independent of η_s.

Proof. After multiplying by $g_s b_\ell^{s-1}$, substitution of $x = a_\ell/b_\ell$ in (41) defines an $\mathbf{F}_\ell$-linear map

$$\overline{M}^+_{D,s} \longrightarrow \mathbf{F}_\ell[f]_{\leq D+(s-1)\delta_\ell+O_s(1)}.$$

The source has dimension $sD+O_s(1)$, while the target has dimension at most $D+(s-1)\delta_\ell+O_s(1)$. Rank–nullity and Lemma 5.2 give (45). Since $g_s(0)b_\ell(0)$ is a unit, vanishing after clearing denominators implies (46). Lift a basis and extend it to an integral basis to obtain a saturated summand. The connection and multiplier are independent of the constant η_s. $\square$

The genuine contraction separates the unique depth-s row from the remaining rows:

$$f^D\langle\lambda, S_{p,s,\eta_s}\rangle = G_{s,\lambda} + q_{0,s,\lambda}H^r J_{p,s} + \eta_s q_{0,s,\lambda}H^r + \sum_{e=1}^{s-1} \widetilde{Q}_{s,e,\lambda}R_{s,e}. \tag{48}$$

Every remaining nonalgebraic row has depth at most $s-1$.

Fix $1 < \xi < m$ and put

$$N^\sharp = \lfloor \xi D \rfloor + C_{\rm sh}, \qquad a_{\ell,D} = v_\ell(L_{N^\sharp}).$$

For each good small prime and each $s \in \mathcal{S}$, choose a complement $K^c_{D,\ell,s}$ and define the block lattice

$$M^{\rm H}_{D,\ell,s} = \ell^{sa_{\ell,D}-1} K_{D,\ell,s} \oplus \ell^{sa_{\ell,D}} K^c_{D,\ell,s}. \tag{49}$$

Together with the unscaled scalar block and the direct sum over s, this is the multiweight small-prime lattice.

Proposition 5.4 (Multiweight small-prime cost)**.** *The lattice* (49) *has integral evaluation through* $f^{\lfloor \xi D \rfloor}$*, and*

$$\sum_{\ell \le N^\sharp} \log[M_{D,\ell} : M^{\rm H}_{D,\ell}] = (\xi Q_{\mathcal{S}} - B_{\mathcal{S}} J_p(\xi))\, D^2 + o(D^2). \tag{50}$$

The residual small-prime evaluation heights are $o(D^2)$*.*

Proof. On $K_{D,\ell,s}$, (47) restores the one missing power of ℓ on the unique depth-s row. Since $sa_{\ell,D} - 1 \ge ea_{\ell,D}$ for $e \le s-1$, every other row is integral as well. The block index is

$$s^2 D a_{\ell,D} - (s-1)\max\{0, D - \lceil c_p \ell \rceil - C_s\} + O_s(a_{\ell,D}).$$

Sum over $s \in \mathcal{S}$. The prime number theorem and partial summation give

$$\sum_{\ell \le \xi D} a_{\ell,D} \log \ell = \xi D + o(D)$$

and

$$\sum_{\ell \le \xi D} \max\{0, D - c_p \ell\} \log \ell = D^2 J_p(\xi) + o(D^2).$$

The finite exceptional set and proper-prime-power residuals contribute only $O_{p,\mathcal{S}}(D \log D) = o(D^2)$. □

6 Multiweight Cartier strictness at large primes

On one fixed flat torsor chart, the normalized simultaneous contraction has the form

$$F_\lambda(f) = H_\lambda(f,u) + \sum_{s \in \mathcal{S}} \sum_{e=1}^{s} Q_{s,e,\lambda}(f,u) R_{s,e}(f), \tag{51}$$

where every multiplier has f-support in $[-C, D+C]$ and fixed fibre degree. Inserting Lemma 3.3 gives

$$F_\lambda = H^\flat_\lambda + \sum_{s \in \mathcal{S}} \sum_{e=1}^{s} \ell^{-e} Q_{s,e,\lambda}(f,u) R_{s,e}(\varphi_\ell(f)), \tag{52}$$

with integral $H^\flat_\lambda$.

Lemma 6.1 (Coefficient-ring Taylor no-backflow)**.** *Let A be an $\mathbf{F}_\ell$-algebra and*

$$\Gamma(f,z) = \sum_{j=0}^{d} f^j P_j(z), \qquad d < \ell, \qquad P_j(z) \in A[[z]].$$

If $\Gamma(f, f^\ell) \in f^n A[[f]]$ *and* $\Delta(f) \in f^{\ell+1} A[[f]]$*, then for every* $t \ge 0$*,*

$$\Delta(f)^t\, \partial_z^t \Gamma(f,z)\Big|_{z=f^\ell} \in f^{n+t} A[[f]]. \tag{53}$$

Proof. Write $P_j(z) = \sum_r p_{j,r} z^r$. Since $0 \le j < \ell$, the exponent $j + \ell r$ uniquely determines (j, r), so no cancellation between different pairs is possible, even when A has zero divisors. A nonzero coefficient therefore satisfies $j + \ell r \ge n$. After t derivatives and multiplication by Δ^t, its order is at least

$$j + \ell(r - t) + t(\ell + 1) = j + \ell r + t \ge n + t.$$

□

Let

$$F_\lambda(f) = c_n f^n + O(f^{n+1}), \qquad c_n \ne 0,$$

be a Bost pivot. Since $n = O_{\mathcal{S}}(D)$ by Corollary 4.3 and $\ell > \xi D$, terms of z-degree at least ℓ in (52) lie beyond the pivot range. Truncate every raw row below z^ℓ and shift by f^C. Put

$$B_{s,e}(f, u, z) = f^C Q_{s,e,\lambda}(f, u) R_{s,e,<\ell}(z). \tag{54}$$

Then

$$\deg_f B_{s,e} < \ell, \qquad \deg_z B_{s,e} < \ell, \qquad B_{s,e}(f, u, 0) = 0.$$

Choose digits

$$B_{s,e} \equiv \sum_{\nu=0}^{e-1} \ell^\nu B_{s,e,\nu} \pmod{\ell^e}. \tag{55}$$

For $1 \le b \le s_*$ define

$$\Gamma_b(f, z) = \sum_{s \in \mathcal{S}} \sum_{e=b}^{s} \overline{B}_{s,e,e-b}(f, u, z). \tag{56}$$

If Π_b denotes the pole-grade-b symbol of the shifted truncated expression, Taylor expansion gives

$$\boxed{\Pi_b(f) = \sum_{t=0}^{s_*-b} \frac{\overline{\Delta}_\ell(f)^t}{t!} \, \partial_z^t \Gamma_{b+t}(f, z)\Big|_{z=f^\ell}.} \tag{57}$$

All factorials are units after excluding the finite set $\ell \le s_*$.

Proposition 6.2 (Multiweight torsor Cartier strictness)**.** *If $v_\ell(c_n) < 0$ and $a = -v_\ell(c_n)$, then*

$$1 \le a \le s_*$$

and there exist

$$-C \le j \le D + C, \qquad r \ge 1, \qquad n = j + \ell r.$$

Consequently

$$\delta_\ell(\psi_D^{(n)}) \le s_* v_\ell \left(\operatorname{lcm}\{\max(n - D - C, 1), \ldots, n + C\}\right) \log \ell. \tag{58}$$

Proof. After truncation, the only negative powers of ℓ are the explicit ℓ^{-e} with $e \le s_*$, so $a \le s_*$. Every coefficient below the pivot order vanishes. Taking pole-grade symbols gives $\Pi_b \in f^{n+C} A[[f]]$. Descending induction on b, using Lemma 6.1 in (57), proves

$$\Gamma_b(f, f^\ell) \in f^{n+C} A[[f]], \qquad \Pi_b \equiv \Gamma_b(f, f^\ell) \pmod{f^{n+C+1}}.$$

At the actual pole grade a, the coefficient of f^{n+C} in Π_a is the nonzero reduction of $\ell^a c_n$. Hence the same coefficient in $\Gamma_a(f, f^\ell)$ is nonzero. Some contributing monomial has

$$n + C = j' + \ell r, \qquad 0 \le j' \le D + 2C, \qquad r \ge 1.$$

Set $j = j' - C$. Then $\ell r = n - j$ lies in the displayed interval. Since $a \le s_*$, (58) follows. □

Proposition 6.3 (Rectangular large-prime cost). *The total positive denominator penalty at primes $\ell > N^\sharp$ satisfies*

$$\mathscr{A}_D^{\mathrm{large}} \le s_* I_\xi^m(\xi) D^2 + o(D^2). \tag{59}$$

Proof. Apply Proposition 6.2 and the CDT packing sum for the $mD + O(1)$ distinct pivots contained in an interval of normalized length $m + o(1)$. Fixed shifts by C and C_{sh} contribute $O(D \log D) = o(D^2)$. □

6.1 The thickness-sensitive Fitting refinement

Let $L_{D,\ell}$ be the integral simultaneous source lattice at a good large prime and

$$L_{D,\ell}^{(n)} = L_{D,\ell} \cap \{\lambda : \mathrm{ord}_f \langle \lambda, S \rangle \ge n\}$$

its saturated vanishing filtration. At a pivot n, write

$$a_{\ell,n} = \max\{0, -v_\ell(c_n)\}.$$

For $1 \le b \le s_*$ define

$$\mathcal{Q}_{b,\ell} = \bigoplus_{\substack{s \in \mathcal{S} \\ s \ge b}} L_{D,s,\ell}^+ / \ell^{s-b+1} L_{D,s,\ell}^+. \tag{60}$$

Its length is

$$\mathrm{length}_{\mathbf{Z}_\ell} \mathcal{Q}_{b,\ell} = \rho_b D + O_{p,\mathcal{S}}(1),$$

with ρ_b as in (8).

Proposition 6.4 (Pole-grade Fitting bound). *For every good large prime ℓ and $1 \le b \le s_*$,*

$$\#\{n \in V_D : a_{\ell,n} \ge b\} \le \rho_b D + O_{p,\mathcal{S}}(1). \tag{61}$$

Proof. Let $U_{b,n}$ be the image of $L_{D,\ell}^{(n)}$ in $\mathcal{Q}_{b,\ell}$. These form a descending chain. Suppose $a_{\ell,n} \ge b$, and choose a primitive lift λ_n of the rank-one pivot quotient. If its image in $\mathcal{Q}_{b,\ell}$ belonged to $U_{b,n+1}$, there would be $\mu \in L_{D,\ell}^{(n+1)}$ such that

$$(\lambda_n - \mu)_s \in \ell^{s-b+1} L_{D,s,\ell}^+ \qquad (s \ge b).$$

The s-block has maximum pole depth s, so these components now have depth at most $b-1$; blocks with $s < b$ already have depth at most $b-1$, and the scalar block has depth zero. Since μ vanishes through the pivot order, $\lambda_n - \mu$ has the same leading coefficient c_n, forcing $v_\ell(c_n) \ge -(b-1)$, a contradiction. Therefore $U_{b,n+1} \subsetneq U_{b,n}$. Each grade-$\ge b$ pivot consumes at least one unit of the total length of $\mathcal{Q}_{b,\ell}$, proving (61). □

Remark 6.5 (Why rank alone is insufficient). One depth-s source direction may successively consume several powers of ℓ. This is why the correct cap is the length $s - b + 1$ per direction, not merely the rank of the deep source. The quotient (60) records exactly that thickness.

Proposition 6.6 (Capped prime-window sum). *For every integer $\rho \ge 1$, the function* (9) *satisfies* (12). *Moreover the pole-grade Fitting bounds imply*

$$\mathscr{A}_D^{\mathrm{large}} \le \left(\sum_{b=1}^{s_*} I_\xi^{m,\rho_b}(\xi) \right) D^2 + o(D^2). \tag{62}$$

Proof. At a prime $\ell = xD$, the usual degree-D multiplier window contains at most $w_m(x)D + o(D)$ pivots. By Proposition 6.4, at most $\rho_b D + o(D)$ of them can have pole grade at least b. Hence the grade-b contribution is bounded by $\min\{w_m(x), \rho_b\}D + o(D)$. Prime-number-theorem summation yields (62).

The function w_m is decreasing and equals ρ at $x = m/(\rho+1)$. If ξ is below that point, the cap contributes $\rho(m/(\rho+1) - \xi)$ before the ordinary tail begins. Evaluating I_ξ^m at $m/(\rho+1)$ gives (12). If the cutoff is already beyond the crossing, the cap is inactive. □

7 Bost–slopes assembly

We record the fixed-bundle bridge needed for literal application of the scalar CDT estimates.

Lemma 7.1 (Adelic fixed-bundle comparison)**.** *Let $\mathcal{V}$ be a fixed rank-m vector bundle on $\mathbf{P}^1_{\mathbf{Q}}$ with fixed adelic models and metrics, and put*

$$N_D = H^0(\mathbf{P}^1, \mathcal{V}(DP + B)).$$

Suppose scalar contraction with a fixed horizontal leaf is injective and its pivot set satisfies a linear zero estimate. Then fixed changes of frame, normalizer, bundle splitting, local model, or scalar row degree change the source degree and summed evaluation heights by $o(D^2)$. In particular,

$$\widehat{\deg} N_D = \frac{m}{2}\mathscr{H} D^2 + o(D^2), \tag{63}$$

and

$$\sum_{n\in V_D}\sum_v h_v(\psi_D^{(n)}) \le \left(-\frac{m^2}{2}\Lambda + m\mathscr{H}\right) D^2 + \mathscr{A}_D + o(D^2). \tag{64}$$

Proof. By Birkhoff–Grothendieck, after a fixed twist $\mathcal{V} \simeq \bigoplus_i \mathcal{O}(a_i)$. The original and transported adelic norms are uniformly equivalent; the determinant distortion is $O(D)$. The scalar arithmetic Hilbert–Samuel formula gives $\frac{1}{2}\mathscr{H}(D + a_i)^2$ on each summand, proving (63). Fixed analytic matrices and normalizers contribute $O(\log D)$ per pivot, hence $O(D \log D) = o(D^2)$ in total. The scalar CDT height estimate contributes $-n\Lambda + D\mathscr{H}$ at each pivot. Since the $mD + O(1)$ pivot orders are distinct and nonnegative,

$$\sum_{n\in V_D} n \ge \frac{1}{2}m^2 D^2 + o(D^2),$$

which proves (64). □

Proof of Theorem 1.1. By Propositions 5.4 and 6.3 and Lemma 7.1, the multiweight Hasse lattice satisfies

$$\widehat{\deg} M_D^{\mathrm{H}} = \left(\frac{m}{2}\mathscr{H} - \xi Q_{\mathcal{S}} + B_{\mathcal{S}} J_p(\xi)\right) D^2 + o(D^2)$$

and

$$\sum_{n\in V_D}\sum_v h_v(\psi_D^{(n)}) \le \left(-\frac{m^2}{2}\Lambda + m\mathscr{H} + s_* I_\xi^m(\xi)\right) D^2 + o(D^2).$$

Bost's slopes inequality gives the first expression no larger than the second. Rearranging yields (6) with (7). □

Proof of Theorem 1.2. Replace Proposition 6.3 by Proposition 6.6. The small-prime lattice and every other Bost term are unchanged. The same rearrangement gives (11). □

8 Common analytic continuation and collision energy

We first close the extra continuation interface required at $p = 13$.

Lemma 8.1 (The 13-adic supersingular annulus)**.** *Put $t = 13 f_{13}$. Then the unique supersingular annulus of $X_0(13)$ is*

$$0 < v_{13}(t) < 1.$$

The wide-open $W_0(13)$ containing the Tate cusp is therefore

$$W_0(13) = \{v_{13}(t) > 0\} = \{|f_{13}|_{13} < 13\}. \tag{65}$$

Proof. By [15, (3.2) and Table 4], Maier's canonical Hauptmodul is $t = t_{13} = 13 f_{13}$ and the degree-14 map to the j-line is

$$j = \frac{(t^2 + 5t + 13)(t^4 + 7t^3 + 20t^2 + 19t + 1)^3}{t}. \tag{66}$$

Deuring's count [14] gives a single supersingular j-invariant in characteristic 13. It is $j = 5$: the curve

$$E : y^2 = x^3 + x + 4$$

has $j(E) = 5$ in $\mathbf{F}_{13}$, and the exact quadratic-character sum

$$\sum_{x \in \mathbf{F}_{13}} \left(\frac{x^3 + x + 4}{13} \right) = 0$$

gives $\#E(\mathbf{F}_{13}) = 14$. Its Frobenius trace is therefore zero, so E is supersingular.

Write $v = v_{13}(t)$. If $v < 0$, the highest-degree terms in the two factors of the numerator of (66) dominate, and $v_{13}(j) = 13v < 0$. If $0 < v < 1$, the terms $5t$ and 1 dominate, so $v_{13}(j) = 0$ and

$$j \equiv 5 \pmod{13}.$$

If $v > 1$, the terms 13 and 1 dominate, and $v_{13}(j) = 1 - v < 0$. Thus the open tube with supersingular reduction is $0 < v < 1$; the loci $v = 0$ and $v = 1$ are its two boundary ends and are not part of the strict wide-open. The cusp ∞ satisfies $v(t) > 1$; adjoining the supersingular annulus to its ordinary component gives $v(t) > 0$, which is (65) because $t = 13 f_{13}$. □

Theorem 8.2 (Common continuation)**.** *For $p \in \{2, 3, 5, 7, 13\}$ and every finite packet $\mathcal{S}$, the simultaneous horizontal leaf extends over the same domains as its individual blocks. At the distinguished p-adic place it extends over*

$$|f_p|_p < p^{12/(p-1)}.$$

At every finite place $\ell \neq p$, under joint rationality it is meromorphic on $|f_p|_\ell < 1$. At the complex place it is meromorphic on every modular template used below.

Proof. Each block has U_p-eigenvalue 1 by Lemma 3.1. Inputs (E1)–(E2) extend it across the Buzzard wide-open $W_0(p)$. For $p = 2, 3$ the explicit radii are the calculations of [6, Lemma 3.1 and the proof of Theorem 3.5]. For $p = 5, 7$ the identification of that wide-open with $|f_p|_p < p^{12/(p-1)}$ is the one used in the underlying single-weight argument. For $p = 13$ it is Lemma 8.1. The BGG operator applies only finitely many Gauss–Manin derivatives and does not shrink the base domain. A finite direct sum has the intersection of the blockwise domains, which here is the same common domain. □

For $0 < Y < 1/p$, put

$$\phi_{p,Y}(z) = f_p(e^{-2\pi Y} z)$$

and

$$C_p(Y) = 4\pi \sum_{\substack{c \geq 1 \\ p \mid c \\ cY < 1}} \frac{\varphi(c)}{c^2} \left(\arccos(cY) - cY\sqrt{1 - c^2 Y^2} \right). \tag{67}$$

The expression $C_p(Y)$ itself makes sense for every prime; Theorem 13.11 below identifies it intrinsically as the Archimedean modular collision term at arbitrary prime level.

Proposition 8.3 (Modular Jensen formula). *For every $p \in \{2,3,5,7,13\}$ one has*

$$\iint_{\mathbb{T}^2} \log|\phi_{p,Y}(z) - \phi_{p,Y}(w)| \, \mathrm{d}m(z)\,\mathrm{d}m(w) = -2\pi Y + C_p(Y). \tag{68}$$

Consequently

$$\Lambda_p(Y) = \frac{12\log p}{p-1} - 2\pi Y, \qquad \mathscr{H}_p(Y) = \Lambda_p(Y) + C_p(Y). \tag{69}$$

Proof. For all five primes f_p is a Hauptmodul. The nonidentity collisions are therefore parametrized by primitive bottom rows (c,d) with $c > 0$ and $p \mid c$. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$,

$$\Im(\gamma(x+iY)) = \frac{Y}{(cx+d)^2 + c^2Y^2}.$$

Jensen contributes $2\pi(\Im(\gamma\tau) - Y)_+$. Unfolding the sum over primitive d and evaluating the elementary integral gives the summand in (67). The leading q contributes $-2\pi Y$, while the eta-product factors have logarithmic mean zero. Adding the distinguished p-adic radius from Theorem 8.2 gives (69). □

Combining Theorem 1.1 and Proposition 8.3, joint rationality forces

$$(m-1)\Lambda_p(Y) - m\tau^{\mathrm{H}}_{p,\mathcal{S}}(\xi) - C_p(Y) \le 0. \tag{70}$$

The branch-and-bound certificates prove strict positivity of the left side for every forbidden aggregate state at $p = 2,3,5,7$; the $p = 13$ finite example below uses a separate elementary bound.

9 Explicit individual and counting applications

9.1 The base individual theorem

For completeness, Table 1 records exact rational witnesses for Theorem 1.7. The margins are rigorous outward interval lower bounds from the single-weight certificate.

Table 1: Single-weight witnesses used to start the counting induction.

p	s	ξ	$\tau_{p,s}(\xi)$	Y	certified lower margin
2	3	22/19	1321/608	16/79	9.745786633966
2	5	29/26	18445/5616	6/49	13.454727090366
2	7	110/101	105585/25856	1/11	15.805524331097
2	9	89/83	10906993/2324000	3/43	17.190005995811
2	11	262/247	464385991/89631360	3/52	17.790824360622
2	13	181/172	144989487/25958240	2/41	17.737379764768
2	15	478/457	16666007411/2810615808	3/71	17.124944575766
2	17	305/293	12175769323/1954839744	3/80	16.023242852853
2	19	758/731	11693630797/1801184000	1/30	14.484310655378
2	21	461/446	536638388131/79764338368	1/33	12.553794492186
2	23	1102/1069	22108077629089/3185347156992	1/36	10.264641765760
2	25	649/631	21299386717285/2985462031552	1/39	7.646580292916
2	27	1510/1471	456079886220493/62372115303680	1/42	4.724811329943
2	29	869/848	14321093408042087/1915475381820000	1/46	1.522247623229
3	3	33/29	2029/928	4/27	6.200294149415
3	5	87/79	112477/34128	7/73	6.941905556192
3	7	55/51	53435/13056	1/15	6.163732078651

p	s	ξ	$\tau_{p,s}(\xi)$	Y	certified lower margin
3	9	267/251	33037241/7028000	1/19	4.332047506141
3	11	393/373	702154669/135354240	3/70	1.679644254174
5	3	11/10	177/80	7/71	2.515479614080
5	5	29/27	12899/3888	1/16	0.131799356827
7	3	33/31	2219/992	1/14	0.436413057785

9.2 The finite branch-and-bound certificate

For each p, list the rational arguments in increasing order as $r_{p,1} < r_{p,2} < \cdots$. The individual theorem gives the first lower bound. At the kth stage, fix a proposed upper bound x for $r_{p,k}$. For each possible sum T of the previous $k-1$ entries, the margin depends on those entries only through

$$m = 1 + T + x, \qquad Q = \sum_{i<k} r_{p,i}^2 + x^2, \qquad B = m - 1 - k, \qquad s_* = x.$$

For fixed T and x, the arithmetic type is increasing in Q. The following majorization lemma therefore reduces the search to one extremal configuration per aggregate state.

Lemma 9.1 (Reverse-greedy square-sum maximizer)**.** *Let $a_1 < \cdots < a_n < x$ be odd integers with prescribed lower bounds and prescribed total sum. Among all such tuples, $\sum a_i^2$ is maximized by allocating the available excess successively to $a_n, a_{n-1}, \ldots$, subject to the gaps $a_i \le a_{i+1} - 2$ and $a_n \le x - 2$.*

Proof. If $i < j$, a_i can be decreased by 2, and a_j increased by 2 without violating the constraints, then the square sum changes by

$$(a_i - 2)^2 + (a_j + 2)^2 - a_i^2 - a_j^2 = 4(a_j - a_i) + 8 > 0.$$

Repeated exchanges force all available excess as far to the right as the constraints allow, which is exactly the reverse-greedy configuration. □

The certificate evaluates (70) for every aggregate state with a fixed rational pair (ξ, Y) at each stage. All algebraic quantities are exact. The values π and $\log p$, square roots, inverse trigonometric functions, and collision sums are enclosed by outward-rounded fixed-point intervals. No binary floating point is used.

The resulting rational-index lower bounds are:

$$\begin{aligned}
p = 2 :&\quad 31, 105, 261, 553, 1069, 1937, > 2401;\\
p = 3 :&\quad 13, 39, 87, 169, 303, 513, 831, 1301, 1983, > 2401;\\
p = 5 :&\quad 7, 15, 29, 51, 81, 127, 189, 275, 391, 547, 751, 1019, 1365, 1809, 2373, > 2401;\\
p = 7 :&\quad 5, 9, 15, 25, 39, 57, 81, 111, 153, 205, 271, 355, 459, 589, 749, 945,\\
&\qquad\qquad 1185, 1477, 1829, 2255, > 2401.
\end{aligned}$$

Here the jth displayed finite number is a certified lower bound for $r_{p,j}$.

Proposition 9.2 (A uniform finite bound at $p = 13$)**.** *Among*

$$\zeta_{13}(3), \zeta_{13}(5), \ldots, \zeta_{13}(2401),$$

at most 139 *values are rational. Hence at least* 1061 *are irrational. In the shorter list ending at $s = 2279$, at least* 1000 *are irrational.*

Table 2: Explicit counting examples; the first four rows are interval-certified.

p	aggregate states	smallest stage margin	rational cap	irrational count	1000 cutoff
2	2,591,454	> 2.507569047577	6	1194	2013
3	4,136,947	> 2.005014256508	9	1191	2019
5	6,889,305	> 0.687480581811	15	1185	2029
7	9,493,807	> 0.388266442986	20	1180	2039
13	–	> 0.008	139	1061	2279

Proof. Suppose that 140 values are rational and take them as $\mathcal{S}$. Then

$$m = 1 + \sum_{s\in\mathcal{S}} s \geq 1 + (3 + 5 + \cdots + 281) = 19881, \qquad s_* \leq 2401,$$

and $Q_{\mathcal{S}} \leq s_*(m-1)$. Drop the favorable Hasse subtraction in (7) and let $\xi \downarrow 1$. Since

$$I_1^m(1) = \left(m - \frac{1}{2}\right) H_{m-1} - (m-1),$$

the terms $s_*(m-1)$ cancel. Since the right-hand side has this finite right limit as $\xi \downarrow 1$, an admissible $\xi > 1$ may be chosen arbitrarily close to 1, and

$$\inf_{\xi>1} \tau^{\mathrm{H}}_{13,\mathcal{S}}(\xi) \leq \frac{2s_*(m-\frac{1}{2})H_{m-1}}{m^2}. \tag{71}$$

The sequence $H_n/(n+1)$ is decreasing for $n \geq 2$, since $H_n > 1$. The classical bound $H_n < \log n + \gamma + 1/(2n)$, together with $\gamma < 289/500$ and $\log 19880 < 10$, gives the exact rational estimate

$$\inf_{\xi>1} \tau^{\mathrm{H}}_{13,\mathcal{S}}(\xi) < \frac{4802 \cdot 10579}{1000 \cdot 19881} < \frac{255523}{100000}.$$

Choose $Y = 1/m$. Since $0 \leq \arccos x - x\sqrt{1-x^2} \leq \pi/2$ and $\varphi(c) \leq c$,

$$C_{13}(1/m) \leq \frac{2\pi^2}{13} H_{\lfloor (m-1)/13 \rfloor}.$$

The functions H_{m-1}/m and $(1 + \log m)/m$ decrease for the present range; the first assertion follows from $H_{m-1} > 1$, and the second by differentiation. Using

$$\log 13 > \frac{25649}{10000}, \qquad \pi < \frac{22}{7}, \qquad H_n \leq 1 + \log n,$$

and again $\log 19881 < 10$, the analytic lower bound is minimized at $m = 19881$ and satisfies the exact rational estimate

$$\begin{aligned}\frac{(m-1)\Lambda_{13}(1/m) - C_{13}(1/m)}{m} &\geq \left(1 - \frac{1}{19881}\right)\left(\frac{25649}{10000} - \frac{44}{7 \cdot 19881}\right) \\ &\quad - \frac{2(22/7)^2}{13 \cdot 19881} \cdot 11 > \frac{256361}{100000}.\end{aligned}$$

The difference between this lower bound and the preceding arithmetic upper bound is greater than $8/1000$, contradicting (70). Thus at most 139 values in the full list are rational. The same proof applies to the prefix ending at 2279, where s_* is smaller; that prefix contains 1139 odd arguments, so at least $1139 - 139 = 1000$ are irrational. $\square$

Proof of Theorem 1.3. The displayed rational-index bounds show that, up to 2401, there can be at most $6, 9, 15, 20$ rational values for $p = 2, 3, 5, 7$, respectively. Subtracting from 1200 gives the first four certified rows of Table 2. For the 1000 cutoffs, there are respectively $1006, 1009, 1014, 1019$ positive odd arguments up to $2013, 2019, 2029, 2039$. The same rational-index lower bounds show that at most $6, 9, 14, 19$ of those shorter lists can be rational. The $p = 13$ row is Proposition 9.2. Hence at least 1000 are irrational in each of the five displayed cases. □

9.3 Certificate files and manifests

The base individual margins in Table 1 are reproduced by `hybrid_rational_interval_certificate.py`; its stored output and manifest are `hybrid_rational_interval_certificate_output.txt` and `hybrid_rational_interval_certificate_sha256.txt`. The manifest records

```
1408c9092d0c34917253c8f520db56853112c58640d15c98d58b76a61a0478f5
  hybrid_rational_interval_certificate.py
9e1d8f74eb198c7e7b0c7420a1e8021d7bad4d2bc8014cc939b6b353111c0da0
  hybrid_rational_interval_certificate_output.txt
```

The multiweight program `multiweight_all_primes_interval_certificate.py` checks 23,111,513 aggregate states. Its stored output is `multiweight_all_primes_interval_certificate_output.txt`. The manifest `multiweight_all_primes_interval_certificate_sha256.txt` records

```
4f9cfbfe9691447fb352a000b360029bdef46f4f66dea7fa4a45f414f247c1f0
  multiweight_all_primes_interval_certificate.py
46b3325cd5a9e6e81b7dfa565361d845e1480aa8f7c3251727b81ca4b583e89b
  multiweight_all_primes_interval_certificate_output.txt
```

The positive-genus exact-capacity bounds in Theorem 1.4 are checked by `allprime_capacity_finite_packet_certificate.py`. The script uses outward-rounded integer fixed-point intervals only; its stored output is `allprime_capacity_finite_packet_certificate_output.txt`. The manifest `allprime_capacity_finite_packet_certificate_sha256.txt` records

```
52e404ad0ea803526a0f682cb0f9e8049ff34d29f32bbd1c590925d421c1168f
  allprime_capacity_finite_packet_certificate.py
3e2a4104ab71582b5ed1873805c2ccdd9f2d867d54a9669fda1a43be6574f75b
  allprime_capacity_finite_packet_certificate_output.txt
```

10 Quantitative density one at genus-zero prime level

Put

$$L_p = \frac{12 \log p}{p - 1}.$$

We now prove Theorem 1.5. Only the rectangular multiweight theorem is needed; in particular, neither the Hasse subtraction nor the Fitting polygon is load-bearing.

Proof of Theorem 1.5. Fix $\beta > 1$. For integers $j \geq 1$, let $\mathcal{S}_j$ be the set of rational odd arguments in the multiplicative shell

$$\beta^j \leq s < \beta^{j+1},$$

and put $k_j = \#\mathcal{S}_j$. Empty shells may be ignored. Set $A = \beta^j$ and form the simultaneous packet attached to $\mathcal{S}_j$. Then

$$k_j A \le m - 1 < k_j \beta A, \qquad s_* < \beta A, \qquad Q_{\mathcal{S}_j} \le s_*(m-1) < \beta A m.$$

Choose $\xi = 1 + 1/\log A$ for A sufficiently large, drop the favorable Hasse term, and use

$$I_\xi^m(\xi) \le I_1^m(1) < m H_{m-1}.$$

The arithmetic type therefore satisfies

$$\tau^{\mathrm{H}}_{p,\mathcal{S}_j}(\xi) \le \frac{2\beta}{k_j}(\xi + H_{m-1}). \tag{72}$$

For the analytic template take $Y = 1/m$. Since

$$0 \le \arccos x - x\sqrt{1-x^2} \le \frac{\pi}{2}$$

and $\varphi(c) \le c$, the collision formula gives

$$C_p(1/m) \le \frac{2\pi^2}{p} H_{\lfloor (m-1)/p \rfloor} = O_p(\log m). \tag{73}$$

Hence the rationality inequality (70) implies

$$\tau^{\mathrm{H}}_{p,\mathcal{S}_j}(\xi) \ge L_p - O_p\left(\frac{\log m}{m}\right). \tag{74}$$

Initially $k_j = O_\beta(A)$, so $m = O_\beta(A^2)$ and $H_{m-1} = O_\beta(\log A)$. For all sufficiently large j, (74) is at least $L_p/2$; combining it with (72) therefore first gives $k_j = O_{p,\beta}(\log A)$. Substitution back into the source-rank bound gives

$$m = O_{p,\beta}(A \log A), \qquad H_{m-1} = \log A + O_{p,\beta}(\log\log A).$$

Consequently

$$k_j \le \left(\frac{2\beta}{L_p} + o(1)\right)\log A = \left(\frac{2\beta \log \beta}{L_p} + o(1)\right) j. \tag{75}$$

If $\beta^J \le X < \beta^{J+1}$, summing (75) over $j \le J$ yields

$$R_p(X) \le \left(\frac{\beta}{L_p \log \beta} + o(1)\right)(\log X)^2.$$

The function $\beta/\log\beta$ has its minimum e at $\beta = e$. This proves (14). There is no hidden uniformity assumption in this passage: for each fixed shell, the finite packet $\mathcal{S}_j$ is fixed while the source degree D tends to infinity; only after the exact multiweight inequality has been obtained do we let j tend to infinity. Packet-dependent exceptional primes and lower-order constants have therefore disappeared before the shell summation. Since the number of positive odd integers at most X is $X/2 + O(1)$, the density assertion follows. □

Remark 10.1 (Why this does not give all but finitely many). The estimate permits an infinite exceptional set with only $O_p((\log X)^2)$ elements up to X. The multiweight mechanism gains rank only from values simultaneously assumed rational, so it cannot by itself exclude an arbitrarily sparse sequence of rational exceptions. This is the precise motivation for the anchor-packet problem.

11 Arbitrary prime level: finite-map restriction of scalars

Fix an arbitrary prime p, put $X_p = X_0(p)_{\mathbf{Q}}$, and let $P = \infty$ be the width-one Tate cusp. Add the finite elliptic locus and every fixed pole to a divisor B disjoint from P.

Lemma 11.1 (Coarse-curve descent at arbitrary prime level)**.** *For every finite packet of odd weights, the bundle*

$$\mathcal{O} \oplus \bigoplus_{s\in\mathcal{S}} \operatorname{Sym}^{s-1}\mathcal{H},$$

its simultaneous Eisenstein–Eichler connection, canonical BGG lifts, horizontal germ, and contraction pairing descend from one fixed neat cover to $X_p \setminus |B|$. After enlarging a fixed exceptional set, all these objects and the source lattices spread over $\mathbf{Z}[\Sigma^{-1}]$, uniformly in D. The descended horizontal leaf has functional rank $m = 1 + \sum_{s\in\mathcal{S}} s$ and independent coordinates over $\mathbf{C}(X_p)$.

Proof. The frame torsor and Gauss–Manin structures are functorial on a fixed neat cover, so their two pullbacks to the fibre square agree and satisfy the cocycle condition. Away from the elliptic points the coarse-moduli morphism still has the residual generic $\{\pm1\}$-inertia; the stack is not literally equal to its coarse curve. However, -1 acts trivially on $\operatorname{Sym}^{s-1}\mathcal{H}$ because $s-1$ is even, and it acts trivially on the scalar summand, connection, BGG operator, and contraction pairing. Coherent bundles with trivial inertia descend along the tame coarse-moduli morphism. Removing the finitely many elliptic points eliminates the additional inertia, so these objects descend to $X_p \setminus |B|$. The BGG lift is characterized intrinsically by its bottom quotient and top-oper derivative and is therefore compatible with descent. Clearing the finitely many coefficients and fixed poles gives a model over $\mathbf{Z}[\Sigma^{-1}]$.

The joint monodromy proof of Theorem 4.1 is genus-independent: each nonzero Eichler cocycle spans the irreducible module $\operatorname{Sym}^{s-1}(\mathbf{C}^2)$, and the distinct modules form a multiplicity-free direct sum. The functional-independence argument of Corollary 4.2 then applies over $\mathbf{C}(X_p)$. $\square$

Let g_p be the genus of X_p. The second cusp $Q = 0$ is rational and distinct from P.

Theorem 11.2 (Finite-map Shidlovsky zero estimate)**.** *There is a finite map*

$$\pi_p : X_p \longrightarrow \mathbf{P}^1_{\mathbf{Q}}$$

unramified at P, of a fixed degree d_p, such that the source

$$M_D = H^0\Big(X_p, \mathcal{E}^{\vee}_{p,\mathcal{S}}(DP + B)\Big)$$

has

$$\#V_D = mD + O_{p,\mathcal{S}}(1) \tag{76}$$

and

$$V_D \subset [0, d_p mD + O_{p,\mathcal{S}}(1)]. \tag{77}$$

One may take $d_p = 1$ when $g_p = 0$ and $d_p \le 2g_p$ when $g_p \ge 1$.

Proof. Assume $g_p \ge 1$ and put $A = (2g_p - 1)Q$. Riemann–Roch gives $\ell(A + P) = \ell(A) + 1$, so there is $x \in L(A + P) \setminus L(A)$. It has a simple pole at P, defines a finite map $\pi_p = x$ of degree at most $2g_p$, and is unramified at P. Thus $t = x^{-1}$ and the Tate parameter q are both uniformizers at P and define the same order function.

Write $\mathcal{F} = \mathcal{E}^{\vee}_{p,\mathcal{S}}(B)$ and $\pi_p^*(\infty) = P + E$. Since $DP \le D\pi_p^*(\infty)$, projection formula gives an injection

$$M_D \hookrightarrow H^0\Big(\mathbf{P}^1, \pi_{p*}\mathcal{F} \otimes \mathcal{O}(D\infty)\Big).$$

The pushforward is a fixed rank-$d_p m$ vector bundle. After one fixed Birkhoff–Grothendieck splitting, normalized contractions are polynomial combinations of degree $D + O(1)$ of $d_p m$ fixed formal functions.

Let $k = \mathbf{Q}(t)$ and $K = \mathbf{Q}(X_p)$, choose a k-basis $y_1, \dots, y_{d_p}$ of K, and extend scalars to $\mathbf{C}$. If $f_1, \dots, f_m$ are the horizontal coordinates, then the functions $y_j f_i$ are independent over $\mathbf{C}(t)$: a relation would be a $\mathbf{C}(X_p)$-linear relation among the f_i. The derivation d/dt extends uniquely to the finite separable extension K/k; after scalar extension, restricting the first-order connection system therefore gives a derivative-stable $d_p m$-dimensional $\mathbf{C}(t)$-space. The sharp Shidlovsky estimate [7, Theorem 3.2.8] bounds every nonzero normalized contraction by $d_p m D + O(1)$. Finally, Riemann–Roch gives $\dim M_D = mD + O(1)$, evaluation is injective by Lemma 11.1, and the nonzero graded quotients of a one-variable vanishing filtration are one-dimensional. This proves both assertions. The genus-zero case uses a degree-one coordinate. □

Remark 11.3 (Why the covering degree enters only the range). The map π_p multiplies the number of auxiliary holonomic functions by d_p, but it does not multiply the polynomial degree D. The genuine source retains rank $mD + O(1)$; only the ambient pivot interval grows to $d_p m D + O(1)$.

12 Intrinsic q-jet Cartier strictness

The genus-zero proof uses a Hauptmodul and a Taylor correction for its coordinate Frobenius. On X_p we instead remain in the width-one Tate parameter. For every auxiliary prime $\ell \neq p$ the exact relation is

$$R_{s,e}(q) - \ell^{-e} R_{s,e}(q^{\ell}) \in \mathbf{Z}_{\ell}[[q]]. \tag{78}$$

Fix an algebraic local frame of every coefficient bundle at P and use the canonical local frame q^{-D} of $\mathcal{O}(DP)$. If $h \in H^0(X_p, \mathcal{W}(DP + B_0))$, its *normalized q-germ* is $q^D h$ after trivializing B_0 at P. One further fixed shift q^C may be used to remove the bounded negative support coming from the torsor chart. Every normalized coefficient and every normalized pivot below is understood in this sense. This convention makes the order estimates independent of an unstated choice of pole-clearing coordinate.

Lemma 12.1 (Global coefficient bundle and jet injectivity)**.** *For a fixed packet there are a fixed vector bundle $\mathcal{W}$, a fixed divisor B_0 disjoint from P, and a constant C such that every coefficient of a normalized torsor multiplier is the q-germ of a section of*

$$H^0(X_p, \mathcal{W}(DP + B_0)).$$

After excluding a fixed finite set of auxiliary primes, every nonzero reduced normalized coefficient has order at most $D + C$. Consequently, for $\ell > D + C$, reduction followed by the ℓ-jet map at P is injective.

Proof. The representation, BGG operators, bounded torsor-monomial degree, algebraic contraction, and frame changes are fixed. Coefficient extraction is therefore an algebraic map from the source bundle to a fixed finite-rank coefficient bundle; poles outside the chosen chart are absorbed in B_0.

Choose a fixed saturated injection $\mathcal{W}(B_0) \hookrightarrow \mathcal{O}(A)^{\oplus r}$ with A disjoint from P. For a nonzero scalar component $h \in H^0(X_p, \mathcal{O}(DP + A))$, write $h = q^{-D} u(q)$ in the chosen local frame. If h has a pole at P, then $\operatorname{ord}_q u < D$; if it is regular at P, the principal-divisor identity gives $\operatorname{ord}_P h \leq \deg A$ and hence $\operatorname{ord}_q u \leq D + \deg A$. Thus every nonzero normalized component has order at most $D + \deg A$. After spreading out and excluding finitely many primes, the injection and the relevant section spaces remain saturated and commute with reduction; the same divisor-degree argument holds on the smooth geometrically integral special fibre. A nonzero section in the kernel of the ℓ-jet map would then have order at least $\ell > D + C$, a contradiction. $\square$

Lemma 12.2 (Exact q-block no-backflow). *Let*

$$\Gamma(q, z) = \sum_{r=0}^{\ell-1} w_r(q) z^r$$

with coefficients in the reduced coefficient space of Lemma 12.1, and suppose $\ell > D + C$. If $\Gamma(q, q^\ell) \neq 0$, then

$$\operatorname{ord}_q \Gamma(q, q^\ell) = j + \ell r_0$$

for some $0 \leq j \leq D + C$ and $0 \leq r_0 < \ell$. If the z-constant term is zero, then $r_0 \geq 1$.

Proof. Let r_0 be the least index with $w_{r_0} \neq 0$. Jet injectivity gives $j = \operatorname{ord}_q w_{r_0} \leq D + C < \ell$. Its first term after substitution occurs strictly before $\ell(r_0 + 1)$, whereas every later z-block begins at least at $\ell(r_0 + 1)$. No later block can cancel it. This is valid over an arbitrary coefficient algebra, including one with zero divisors. $\square$

Proposition 12.3 (General-curve q-jet Cartier strictness). *There are a fixed exceptional set and a constant C such that, for $D \gg 1$ and every good auxiliary prime satisfying*

$$\ell > D + C, \qquad \ell > s_*, \qquad \ell \neq p,$$

the following holds. If a normalized Bost pivot

$$F_\lambda(q) = c_n q^n + O(q^{n+1}), \qquad c_n \neq 0,$$

has $v_\ell(c_n) < 0$ and $a = -v_\ell(c_n)$, then

$$1 \leq a \leq s_* \tag{79}$$

and

$$n = j + \ell r, \qquad -C \leq j \leq D + C, \qquad r \geq 1. \tag{80}$$

Consequently

$$\delta_\ell(\psi_D^{(n)}) \leq s_* v_\ell(\operatorname{lcm}\{\max(n - D - C, 1), \ldots, n + C\}) \log \ell. \tag{81}$$

Proof. Insert (78) into the genuine torsor decomposition. By Theorem 11.2, $n = O(D)$; for $\ell > D + C$ and $D \gg 1$ one has $n + C < \ell^2$. Terms of raw z-degree at least ℓ therefore lie beyond the pivot. The coefficients below z^ℓ are ℓ-integral, so the only negative powers are the explicit ℓ^{-e} with $e \leq s_*$, proving (79).

Digitize each complete truncated product, coefficientwise in the finite free global coefficient lattice. For pole grade b let $\Gamma_b(q, z)$ be the sum of the corresponding digits. Because the substitution is exactly $q \mapsto q^\ell$, its associated-graded symbol is simply

$$\Pi_b(q) = \Gamma_b(q, q^\ell);$$

there are no Taylor correction terms. At the actual pole grade a, the q^n-coefficient is the nonzero reduction of $\ell^a c_n$, and every lower coefficient vanishes. Thus $\operatorname{ord}_q \Gamma_a(q, q^\ell) = n$. The raw rows have zero constant term, so Lemma 12.2 gives (80). The positive integer $n - j = \ell r$ lies in the stated interval, and $a \leq s_*$ gives (81). $\square$

For $d, m \geq 1$ and $1 < \xi < dm$, put

$$G_{d,m}(\xi) = \int_{\xi}^{dm} \min\left\{1, \frac{d}{x}\right\} \, \mathrm{d}x = \begin{cases} d - \xi + d\log m, & 1 < \xi < d, \\ d\log\left(\dfrac{dm}{\xi}\right), & d \leq \xi < dm. \end{cases} \tag{82}$$

Proposition 12.4 (General-curve large-prime cost)**.** *For the unmodified source at primes $\ell > \xi D$,*

$$\mathscr{A}_D^{\mathrm{large}} \leq s_* m G_{d_p,m}(\xi) D^2 + o(D^2). \tag{83}$$

Proof. A prime $\ell = xD$ can affect only pivots lying in degree-D windows ending at positive multiples of ℓ. By (77) there are at most $d_p m/x + O(1)$ such windows, while the total number of pivots is $mD + O(1)$. Hence the vulnerable-pivot count is at most $mD\min\{1, d_p/x\} + o(D)$. Apply Proposition 12.3 and the prime number theorem, and integrate over x. □

13 Positive capacitary width and normalization-robust slopes

The arbitrary-prime density theorem needs only a positive tangent-degree term whose coefficient is quadratic in the source rank. It does not require an exact Bost–Charles self-intersection normalization. We therefore work directly with finite-order scalar capacitary estimates and keep every normalization-dependent term at size $O_p(mD^2)$.

For a place v, let (Ω_v, o_v) be a pointed analytic curve and let

$$\iota_v : (\Omega_v, o_v) \longrightarrow (X_p^{\mathrm{an}}, P)$$

be nonconstant and unramified at o_v. Transporting the capacitary norm on $T_{o_v}\Omega_v$ through $D\iota_v(o_v)$ gives a norm $\|\cdot\|_{v,\circ}$ on $T_P X_p$. Put

$$\Lambda_p^{\circ} = \widehat{\deg}(T_P X_p, (\|\cdot\|_{v,\circ})_v). \tag{84}$$

Lemma 13.1 (Strict capacitary collar)**.** *Let K be a non-Archimedean local field, let X/K be a smooth projective curve, and let $P \in X(K)$ extend to a section meeting the special fibre of a normal projective model at a smooth point of an irreducible component C_P. Let Ω_0 be the integral residue disc at P. Let Ω_1 be a connected wide-open containing P for which, on an adapted model, the continuation-side vertical set contains C_P. Then*

$$\|\cdot\|_{P,\Omega_1}^{\mathrm{cap}} < \|\cdot\|_{P,\Omega_0}^{\mathrm{cap}}$$

on every nonzero vector of $T_P X$.

Proof. Let $U_1 = X^{\mathrm{an}} \setminus \Omega_1$. On an adapted normal model, write the Bost–Chambert-Loir extension of the effective divisor P as

$$\mathscr{P}_{\Omega_1} = \mathscr{P} + V, \qquad V = \sum_C c_C C.$$

The divisor V is the unique solution of the graph-Dirichlet problem in [3, Proposition 5.1]. More is proved in the proof of [3, Proposition 5.6]: if the effective divisor meets every connected component of the continuation domain, then

$$c_C = 0 \quad \text{on the excluded side}, \qquad c_C > 0 \quad \text{on the continuation side}.$$

Here Ω_1 is connected and contains P, while C_P lies on its continuation side. Hence

$$c_{C_P} > 0.$$

The strict norm comparison is local at the intersection of the section with C_P. After clearing the denominator of c_{C_P}, choose local parameters q, ϖ with $\mathscr{P} = (q)$ and $C_P = (\varpi)$. Relative to the integral model, a local model frame of $\mathcal{O}(\mathscr{P} + c_{C_P} C_P)$ is $\varpi^{-c_{C_P}} q^{-1}$. Under adjunction, q^{-1} maps to $\partial/\partial q$, and therefore

$$\left\| \frac{\partial}{\partial q} \right\|_{P,\Omega_1}^{\mathrm{cap}} = |\varpi|^{c_{C_P}} \left\| \frac{\partial}{\partial q} \right\|_{P}^{\mathrm{int}} < \left\| \frac{\partial}{\partial q} \right\|_{P}^{\mathrm{int}}. \tag{85}$$

The same calculation after a finite extension, followed by the usual normalization of local degrees, gives the identical inequality when the coefficients are merely rational. Finally, [3, Example 5.11] identifies the integral tangent norm with the capacitary norm of the bare residue disc Ω_0. This proves the claim. □

Proposition 13.2 (Positive common capacitary width)**.** *For every fixed prime p, there is an adelic family $\mathbf{\Omega}_p^\circ = ((\Omega_v, o_v, \iota_v))_v$ such that:*

(i) *every finite simultaneous Eisenstein–Eichler leaf continues along ι_v for every place v;*

(ii) *at all but finitely many finite places, Ω_v is the integral Tate residue disc and ι_v is its inclusion;*

(iii) *the tangent degree in (84) is strictly positive:*

$$\Lambda_p^\circ > 0;$$

(iv) *at the distinguished p-adic place there is a larger continuation wide-open Ω_p^+ such that the affinoid closure of Ω_p° is compactly contained in Ω_p^+. Moreover one may choose a rational function u_p with polar divisor $a_p P$, $a_p \geq 1$, for which*

$$\Omega_p^\circ = \{|u_p|_p > 1\}, \qquad \{|u_p|_p \geq 1\} \Subset \Omega_p^+ \Subset W_0(p). \tag{86}$$

At every Archimedean place the image of the closed pointed domain is likewise contained in a slightly larger continuation domain.

The domains and Λ_p° depend only on p, not on the packet of weights.

Proof. Work first on a fixed sufficiently fine tame level. Buzzard's wide-open $W_0(p)$ is the continuation region through the Tate component, and [5, Theorem 5.2] extends every finite-slope overconvergent eigenform under consideration across it. Each specialization used here has U_p-eigenvalue one, and applying the algebraic BGG operator does not shrink the base domain.

Choose descent-invariant cuts strictly inside all supersingular boundary annuli and pass to the quotient on the coarse curve. The resulting connected wide-open $\Omega_p^\circ \Subset W_0(p)$ contains the tube of the coarse component through P and a positive annular collar at every boundary end. Choose a second set of cuts farther out in the same annuli; it gives a larger connected wide-open Ω_p^+ with $\overline{\Omega_p^\circ} \Subset \Omega_p^+ \Subset W_0(p)$. The descended leaf continues on Ω_p^+ for every weight. By Lemma 13.1, the capacitary tangent metric of Ω_p° has a strict gain over the integral Tate metric:

$$\lambda_p = -\log \frac{\|t\|_{p,\Omega_p^\circ}^{\mathrm{cap}}}{\|t\|_{p,\mathrm{int}}} > 0.$$

Applying the collar lemma after descent avoids any cusp-width or ramification normalization from the auxiliary tame cover.

The complement of Ω_p° is an affinoid meeting every connected component. By [3, Proposition 5.6], it is a rational lemniscate: there is a rational function u_p, with polar divisor $a_p P$, such that its complement is $\{|u_p|_p \le 1\}$. Because the cuts defining Ω_p° were chosen strictly inside those defining Ω_p^+, the affinoid closure $\{|u_p|_p \ge 1\}$ is compactly contained in Ω_p^+. This proves (86).

At every finite place different from p, use the integral Tate residue disc. At the complex place use the pointed unit disc with

$$\iota_\infty(z) : q = e^{-2\pi Y} z.$$

Choose it with a slightly larger concentric disc still contained in the q-coordinate continuation region. Its transported tangent norm contributes $-2\pi Y$ to the Arakelov degree. Choose $Y > 0$ with $2\pi Y < \lambda_p$. The width-one Tate parameter q is an integral parameter on the fixed cusp model, so $\partial/\partial q$ generates the reference tangent lattice at every finite place; together with the unit complex q-disc this reference line has degree zero. Consequently

$$\Lambda_p^\circ \ge \lambda_p - 2\pi Y > 0.$$

Every domain and every larger continuation neighborhood depends only on p, and a finite direct sum of blocks continues on these same domains. □

Put

$$L = \mathcal{O}_{X_p}(P), \qquad \mathcal{F}_{\mathcal{S}} = \mathcal{E}^\vee_{p,\mathcal{S}}(B), \qquad M_D = H^0(X_p, \mathcal{F}_{\mathcal{S}} \otimes L^{\otimes D}).$$

The support of B is disjoint from P and is fixed once the packet is fixed. On each chosen analytic domain, $\mathcal{O}(B)$ is treated as a fixed line-bundle twist: after a fixed power and finitely many residue classes it has a nonvanishing analytic frame. This contributes only packet-dependent lower-order constants. Fix an adelic model of L and a reference adelic norm $\|\cdot\|_{0,v}$ on $T_P X_p$; at every finite place take the reference tangent norm to be the integral Tate norm. Riemann–Roch gives

$$r_D := \operatorname{rank} M_D = mD + O_{p,\mathcal{S}}(1). \tag{87}$$

For the arbitrary-prime argument, choose the reference adelic source lattice blockwise from fixed algebraic presentations as in the next lemma. All good prime coefficient lattices used in the Cartier argument are obtained from the same presentations; enlarging the packet-dependent exceptional set therefore introduces no second, incompatible source model.

Lemma 13.3 (Fixed-twist source and boundary comparison)**.** *There are constants $A_p, B_p \ge 0$, depending only on p, with the following property. For every fixed packet $\mathcal{S}$ there is a constant $C_{p,\mathcal{S}}$ such that*

$$\widehat{\deg} M_D \ge -A_p m D^2 + o_{p,\mathcal{S}}(D^2), \tag{88}$$

and, at the distinguished p-adic and complex domains,

$$\log \|\langle \lambda, S_{p,\mathcal{S}}\rangle\|_{\partial\Omega_v} \le B_p D + C_{p,\mathcal{S}} \log(D+2) + \log\|\lambda\|_v. \tag{89}$$

The boundary estimate remains valid after replacing any finite local source lattice by a sublattice. The resulting loss of source degree is not hidden in (88)*; it is accounted for by its lattice index.*

Proof. Fix once and for all an adelic model of the scalar pair (X_p, L). After replacing L by a fixed positive power and treating the finitely many residue classes as fixed twists, the graded section algebra

$$R_p = \bigoplus_{e \ge 0} H^0(X_p, L^e)$$

is generated by finitely many homogeneous scalar sections. Choose integral lifts of these generators outside one fixed set of places. In weighted degree e, the resulting monomials give a surjection from a free adelic module P_e onto $E_e = H^0(X_p, L^e)$. Equip E_e with the quotient lattice and quotient norm. The number of weighted monomials is polynomial in e, every monomial has Arakelov degree bounded below by $-A_pe - O_p(1)$, and its norm on each of the two fixed analytic boundaries is at most $\exp(B_pe + O_p(1))$. Consequently

$$\widehat{\mu}_{\min}(E_e) \geq -A_pe - O_p(\log(e+2)), \qquad \|E_e \longrightarrow H^0(\partial\Omega_v, L^e)\|_{\mathrm{op}} \leq \exp(B_pe + O_p(\log(e+2))). \tag{90}$$

The first estimate is the finite-generation estimate used in [3, Theorem 6.1, (6.8)]; the monomial presentation also makes the second estimate literal. Replacing these quotient norms by any fixed standard model or L^2 norm changes only the constants A_p, B_p.

Because L is ample, for the fixed bundle $\mathcal{F}_{\mathcal{S}}$ there are fixed integers $a_1, \dots, a_u$ and a sheaf surjection

$$\bigoplus_{j=1}^{u} L^{-a_j} \longrightarrow \mathcal{F}_{\mathcal{S}}.$$

After tensoring by L^D, Serre vanishing for the fixed kernel gives a surjection

$$\bigoplus_{j=1}^{u} E_{D-a_j} \longrightarrow M_D \tag{91}$$

for $D \gg_{p,\mathcal{S}} 1$. Spread this one presentation out and use its quotient lattice and quotient norms. It is chosen separately on the scalar block and on each positive block, so it respects the direct-sum source used in Equation (95). After inverting a finite packet-dependent set, relative Serre vanishing and cohomology-and-base-change identify this quotient lattice with the corresponding integral section lattice. At the remaining places we retain the presentation lattice itself. Thus the same lattice is used in the source-degree, coefficient, and Cartier calculations.

A quotient with its quotient norm has minimum slope no smaller than that of its source. Applying (90) to (91), the fixed shifts a_j, the fixed number u of summands, and the fixed presentation maps contribute only $O_{p,\mathcal{S}}(\log(D+2))$. Hence

$$\widehat{\mu}_{\min}(M_D) \geq -A_pD - O_{p,\mathcal{S}}(\log(D+2)).$$

Multiplication by (87) proves (88). In particular, the coefficient of D is a scalar constant depending only on p; all packet dependence is lower order for a packet fixed before $D \to \infty$.

Finally compose (91) with restriction to a boundary and contraction with the horizontal leaf. The scalar restriction norm is bounded by the second inequality in (90). The finitely many presentation maps, the simultaneous leaf, and a fixed analytic frame for $\mathcal{O}(B)$ have bounded norms depending on the packet but not on D. The finite direct sum contributes at most another packet-dependent constant. Passing to the quotient therefore gives (89). If a finite source lattice is replaced by a sublattice, its gauge norm increases, so the operator-norm upper bound can only improve. □

Let V_D be the rational pivot set of the vanishing filtration and let $G_{D,n}$ be the saturated rank-one quotient at a pivot n, with its quotient adelic norms. Denote its evaluation map by $\psi_D^{(n)}$. For a finite place $v \neq p$, equip the graded target with its integral model and set $\delta_{v,D,n} = \max\{0, \log \|\psi_D^{(n)}\|_v\}$. Thus $\delta_{v,D,n}$ is exactly the positive excess local height at v. Put

$$\delta_{D,n} = \sum_{\substack{v \nmid \infty \\ v \neq p}} \delta_{v,D,n}.$$

The distinguished p-adic operator norm is controlled directly by capacity and is not counted again in $\delta_{D,n}$.

Lemma 13.4 (Finite-order scalar capacitary jet estimate)**.** *There is a constant $B'_p \geq 0$, depending only on p, such that, for every fixed packet, there is a constant $C'_{p,\mathcal{S}}$ satisfying*

$$\widehat{\deg} G_{D,n} \leq B'_p D - n\Lambda_p^\circ + \delta_{D,n} + C'_{p,\mathcal{S}} \log(D+2) \tag{92}$$

for every pivot n and all sufficiently large D. The same estimate holds for a finite adelic sublattice of the reference source, with its induced vanishing filtration and quotient norms.

Proof. The vector bundle is removed before the analytic estimate is applied. Contraction sends λ to an analytic section of $L^D \otimes \mathcal{O}(B)$; after using the fixed frame of the second factor it is a scalar analytic section of L^D. Thus the only vector-bundle input is the bounded contraction operator in Lemma 13.3.

At the p-adic place use the finite-order inequality derived inside the proof of [3, Proposition 5.15], not merely its limiting canonical-norm conclusion. Before the limiting step, that proof uses only a trivialization of a fixed power of L and the maximum principle for the resulting scalar analytic function. It does not use that the section comes from a global scalar linear system, and therefore applies verbatim to the scalar section obtained after contraction. On a rational inner lemniscate a fixed power of L is trivial. Taking D first in the corresponding arithmetic progression and treating the remaining residue classes as fixed twists, the calculation bounds the nth divided jet by the boundary norm times

$$\|t\|_{p,\circ}^n \exp(O_p(D)),$$

relative to a tangent vector t. At the complex place the same assertion is ordinary Cauchy on the chosen q-disc. Combining these estimates with (89) gives

$$\sum_{v\in\{p,\infty\}} \log \|\psi_D^{(n)}\|_v \leq B_p D + n \sum_{v\in\{p,\infty\}} \log \frac{\|t\|_{v,\circ}}{\|t\|_{0,v}} + O_{p,\mathcal{S}}(\log(D+2)). \tag{93}$$

The estimate is valid for the quotient $G_{D,n}$ because its quotient norm is the infimum of the norms of its lifts.

At every remaining finite place one has $\log \|\psi_D^{(n)}\|_v \leq \delta_{v,D,n}$ by definition. The coherent models used to define these quantities are the same models used in the Cartier calculation below; fixed packet denominators are therefore included in the finite exceptional-place term there, rather than in an unrecorded analytic constant. Since the capacitary and reference tangent norms agree at all finite places other than p, summing all places yields

$$h(\psi_D^{(n)}) \leq B_p D - n\big(\Lambda_p^\circ - \widehat{\deg}(\overline{T}_0)\big) + \delta_{D,n} + O_{p,\mathcal{S}}(\log(D+2)). \tag{94}$$

The graded target is

$$\overline{L}_P^D \otimes \overline{T}_0^{\vee\otimes n} \otimes \overline{C}_{\mathcal{S}},$$

where $\overline{C}_{\mathcal{S}}$ is fixed with the packet. The rank-one degree identity, in the normalization of [3, Theorem 6.1, (6.7)], gives

$$\widehat{\deg} G_{D,n} = D\widehat{\deg}(\overline{L}_P) - n\widehat{\deg}(\overline{T}_0) + h(\psi_D^{(n)}) + O_{p,\mathcal{S}}(1).$$

Substituting (94) cancels the arbitrary reference tangent norm and proves (92). This is only a formal change of reference metric; it is not the target-normal Θ-cancellation proved in Theorem 13.7 and Section A. Passing to a finite sublattice increases its gauge norm and therefore cannot increase the analytic operator norm, proving the final assertion. □

Fix $1 < \xi < d_p m$ and put

$$N = \lfloor \xi D \rfloor , \qquad L_N = \mathrm{lcm}(1, \dots, N).$$

Using the blockwise reference lattices above, define

$$M_D^{\mathrm{hyb}} = M_D^0 \oplus \bigoplus_{s \in \mathcal{S}} L_N^s M_{D,s}^+. \tag{95}$$

At the finitely many packet-dependent bad places use the corresponding integral direct-sum presentation lattice. We use this presentation lattice throughout. If it is compared with, or saturated inside, any fixed geometric model, the quotient is killed by one packet-dependent integer because the presentation is fixed; since the rank is $O_{p,\mathcal{S}}(D)$, the resulting logarithmic index is $O_{p,\mathcal{S}}(D) = o(D^2)$.

Proposition 13.5 (Compatibility with the hybrid finite-prime lattices)**.** *For every fixed packet,*

$$\widehat{\deg} M_D^{\mathrm{hyb}} \geq -A_p m D^2 - \xi Q_{\mathcal{S}} D^2 + o_{p,\mathcal{S}}(D^2), \tag{96}$$

and

$$\sum_{n \in V_D} \delta_{D,n} \leq s_* m G_{d_p,m}(\xi) D^2 + o_{p,\mathcal{S}}(D^2). \tag{97}$$

The scalar capacitary estimate (92) remains valid for the quotient norms induced by M_D^{hyb}.

Proof. The positive weight-s block has rank $sD + O_{p,s}(1)$. Scaling it by L_N^s therefore has logarithmic index

$$s^2 D \log L_N + O_{p,s}(\log L_N).$$

Since $\log L_N = N + o(N)$, summing over the blocks and using (88) proves (96). The fixed bad-place and saturation adjustments are $o(D^2)$.

We next bound the finite local heights away from p. By (77) and the fixed multiplier shifts, every raw coefficient index entering a pivot is at most $K_{p,\mathcal{S}} D$ for a fixed constant $K_{p,\mathcal{S}}$. For a prime $\ell \leq N$, $\ell \neq p$, raw depth and multiplication by L_N^s remove the full ℓ-denominator unless $v_\ell(k) > v_\ell(L_N)$ for one of these indices k. Up to the fixed factor s_*, the residual valuation at one pivot is bounded by

$$\begin{aligned} &\sum_{\substack{\ell \leq N \\ \ell \neq p}} \left(v_\ell(L_{\lfloor K_{p,\mathcal{S}} D \rfloor}) - v_\ell(L_N)\right)_+ \log \ell \\ &\qquad \leq \sum_{\substack{a \geq 2 \\ N < \ell^a \leq K_{p,\mathcal{S}} D}} \log \ell = O_{p,\mathcal{S}}(D^{1/2} \log D). \end{aligned}$$

Only proper prime powers occur because $N > D$ for large D. Since $\# V_D = O_{p,\mathcal{S}}(D)$, all residual small-prime heights total $O_{p,\mathcal{S}}(D^{3/2} \log D) = o(D^2)$. The distinguished place p is already controlled by the capacitary Cauchy estimate with the hybrid source norm and is not included in $\delta_{D,n}$. At each of the finitely many other bad primes ℓ, the raw-depth comparison is uniform in D. Indeed,

$$\left(v_\ell(L_{\lfloor K_{p,\mathcal{S}} D \rfloor}) - v_\ell(L_N)\right)_+ = O_{p,\mathcal{S},\xi}(1),$$

because the ratio of the two cutoffs is fixed. The frame, presentation, and torsor denominators likewise have bounded ℓ-valuation. Hence every fixed bad place contributes $O_{p,\mathcal{S},\xi}(1)$ per pivot, and all such places together contribute $O_{p,\mathcal{S},\xi}(D) = o(D^2)$. The separate saturation adjustments in the source degree remain $O_{p,\mathcal{S}}(D \log(D+2)) = o(D^2)$.

For $\ell > N$, the hybrid lattice equals the reference lattice. Since $\xi > 1$, for the fixed packet and large D one has $\ell > D + C$ and $\ell > s_*$. Thus Propositions 12.3 and 12.4 applies; the at most one prime in the harmless floor interval between N and ξD contributes $o(D^2)$. This proves (97).

Finally, $M_D^{\text{hyb}} \subset M_D$ at every finite place. Its gauge norm is therefore larger, and the analytic operator-norm estimates for the reference source can only improve. The vanishing filtration uses the same rational subspaces and the quotient norms induced by this hybrid lattice, so the finite-height and capacitary calculations refer to the same rank-one quotients. This proves the compatibility assertion. ☐

Theorem 13.6 (Normalization-robust general-curve slopes)**.** *For every fixed prime p there are constants*

$$\Lambda_p^\circ > 0, \qquad C_p > 0,$$

depending only on p, such that the following holds. If a fixed finite packet $\mathcal{S}$ is jointly rational, then for every $1 < \xi < d_p m$,

$$\tau_{p,\mathcal{S}}^{\text{curve}}(\xi) = \frac{2}{m^2}\left[\xi Q_{\mathcal{S}} + s_* m G_{d_p,m}(\xi)\right] \tag{98}$$

satisfies

$$\boxed{\tau_{p,\mathcal{S}}^{\text{curve}}(\xi) \geq \Lambda_p^\circ - \frac{C_p}{m}.} \tag{99}$$

The constants are independent of the packet.

Proof. Use the vanishing filtration on M_D^{hyb}. Functional independence makes scalar contraction injective, so the filtration has exactly $r_D = mD + O_{p,\mathcal{S}}(1)$ saturated rank-one quotients. Their Arakelov degrees add:

$$\widehat{\deg} M_D^{\text{hyb}} = \sum_{n \in V_D} \widehat{\deg} G_{D,n}.$$

The pivot orders are distinct nonnegative integers, whence

$$\sum_{n \in V_D} n \geq \frac{r_D(r_D - 1)}{2} = \frac{m^2}{2} D^2 + o_{p,\mathcal{S}}(D^2). \tag{100}$$

Summing (92), using (97), and observing that the packet-dependent $O(\log D)$ term at each of $O(D)$ pivots is $o(D^2)$, gives

$$\widehat{\deg} M_D^{\text{hyb}} \leq B_p' m D^2 - \frac{m^2}{2} \Lambda_p^\circ D^2 + s_* m G_{d_p,m}(\xi) D^2 + o_{p,\mathcal{S}}(D^2).$$

Compare with (96), divide by D^2, and let $D \to \infty$. Then

$$\frac{m^2}{2} \Lambda_p^\circ \leq (A_p + B_p')m + \xi Q_{\mathcal{S}} + s_* m G_{d_p,m}(\xi).$$

Multiplication by $2/m^2$ proves (99) with $C_p = 2(A_p + B_p')$.

Every packet-dependent presentation, frame, bad-place, and boundary constant has disappeared into $o_{p,\mathcal{S}}(D^2)$ before the packet is varied. This is why C_p is genuinely packet-independent, despite the absence of any uniform exceptional-prime set as the packet varies. ☐

The following theorem is the audited replacement for the former exact-normalization conjecture. It proves the required source and jet formulas for a canonical mixed adelic metric on X_p. Only the associated capacitary tangent line is realized as a Bost–Charles normal line; no whole-divisor direct-image map is part of the statement.

Theorem 13.7 (Exact mixed capacitary normalization and target-normal cancellation)**.** *Fix a prime p and the continuation datum $\boldsymbol{\alpha} = \boldsymbol{\Omega}_p^\circ$ of Proposition 13.2. There are a normal arithmetic model $\mathscr{X}_\alpha$ of X_p and a semipositive, nef, and big adelic Arakelov $\mathbf{Q}$-divisor $\widehat{P}_\alpha$ extending P. Put*

$$\overline{L}_\alpha = \mathcal{O}(\widehat{P}_\alpha), \qquad \Theta_\alpha = \widehat{\deg} P^*\overline{L}_\alpha, \qquad \mathscr{H}_\alpha = \widehat{P}_\alpha^2, \qquad \Lambda_\alpha = \widehat{\deg}\overline{T}_\alpha > 0, \tag{101}$$

where $\overline{T}_\alpha$ is the source capacitary tangent line transported to $T_P X_p$. These quantities depend only on p and on the fixed continuation datum.

Let $\mathcal{S}$ be a fixed jointly rational packet, and give the fixed coefficient bundle $\mathcal{F}_\mathcal{S}$ any fixed dominated adelic metric compatible with its integral models. Equip

$$M_D = H^0(X_p, \mathcal{F}_\mathcal{S} \otimes \mathcal{O}(DP))$$

with the induced adelic supremum norms, replacing each Archimedean norm by its John ellipsoid norm. Then

$$\boxed{\widehat{\deg}\overline{M}_D^\alpha = \frac{m}{2}\mathscr{H}_\alpha D^2 + o_{p,\mathcal{S}}(D^2).} \tag{102}$$

The leading term is independent of the fixed metric on $\mathcal{F}_\mathcal{S}$.

For a pivot n, let $\overline{G}_{D,n}^\alpha$ be the saturated rank-one source quotient and let

$$\overline{J}_{D,n}^\alpha = P^*\overline{L}_\alpha^D \otimes \overline{T}_\alpha^{\vee\otimes n} \otimes \overline{C}_\mathcal{S}$$

be its natural jet target, where $\overline{C}_\mathcal{S}$ is fixed. Denote by $h_\alpha^{\mathrm{nat}}(\psi_D^{(n)})$ the global operator height with this target metric. After scalarizing the target by a nonzero rational vector, denote the resulting height by $h_\alpha^{\mathrm{sc}}(\psi_D^{(n)})$; it is independent of the chosen vector by the product formula. For every $C > 0$, uniformly for all pivots $0 \le n \le CD$,

$$\begin{aligned}
\widehat{\deg}\overline{J}_{D,n}^\alpha &= D\Theta_\alpha - n\Lambda_\alpha + O_{p,\mathcal{S}}(1), && (103)\\
h_\alpha^{\mathrm{nat}}(\psi_D^{(n)}) &\le D(\mathscr{H}_\alpha - \Theta_\alpha) + \delta_{D,n} + O_{C,p,\mathcal{S}}(1), && (104)\\
h_\alpha^{\mathrm{sc}}(\psi_D^{(n)}) &\le D\mathscr{H}_\alpha - n\Lambda_\alpha + \delta_{D,n} + O_{C,p,\mathcal{S}}(1), && (105)\\
\widehat{\deg}\overline{G}_{D,n}^\alpha &\le D\mathscr{H}_\alpha - n\Lambda_\alpha + \delta_{D,n} + O_{C,p,\mathcal{S}}(1). && (106)
\end{aligned}$$

Thus the $D\Theta_\alpha$ contribution of the natural target cancels exactly with the $-D\Theta_\alpha$ part of the natural evaluation estimate. In particular,

$$\sum_{n \in V_D} \widehat{\deg}\overline{G}_{D,n}^\alpha \le \left(-\frac{m^2}{2}\Lambda_\alpha + m\mathscr{H}_\alpha\right) D^2 + \mathscr{A}_D + o_{p,\mathcal{S}}(D^2). \tag{107}$$

For the hybrid source of (95), endowed with these metrics,

$$\widehat{\deg}\overline{M}_D^{\alpha,\mathrm{hyb}} = \left(\frac{m}{2}\mathscr{H}_\alpha - \xi Q_\mathcal{S}\right) D^2 + o_{p,\mathcal{S}}(D^2). \tag{108}$$

Therefore joint rationality forces the exact general-curve inequality

$$\boxed{\tau_{p,\mathcal{S}}^{\mathrm{curve}}(\xi) \ge \Lambda_\alpha - \frac{\mathscr{H}_\alpha}{m}.} \tag{109}$$

All formulas are invariant under finite base change after normalized Arakelov degrees are divided by the extension degree.

Proof. The canonical mixed divisor and its nefness and bigness are constructed in Proposition A.3. The denominator-cleared realization of the capacitary tangent line is Proposition A.4; the base-change statement is Lemma A.2. The determinant source formula is Proposition A.6, and the uniform jet estimate and exact $D\Theta_\alpha$ cancellation are Proposition A.10. Finally, Proposition A.12 gives (108), (107), and the slopes assembly (109). □

Remark 13.8 (Exact versus robust normalization). The exact theorem strengthens but does not replace Theorem 13.6. The normalization-robust theorem still gives an independent qualitative proof that $R_p(X) = O_p((\log X)^2)$. The explicit leading coefficient in Theorem 1.6, however, uses the exact modular capacity pair proved below. The exact theorem removes the coarse source and boundary constants A_p and B'_p and replaces their contribution by the intrinsic self-intersection $\mathscr{H}_\alpha/m$.

Remark 13.9 (Scope of the realization bridge). The bridge proved below realizes the full metrized capacitary tangent line as the normal line of a pseudoconcave formal-analytic surface, after clearing rational finite-place exponents. Matching this normal line does not determine the entire Bost–Chambert-Loir finite vertical divisor and does not by itself produce a Bost–Charles meromorphic map to $\mathscr{X}_\alpha$. Neither stronger assertion is used. The exact source and jet formulas are proved directly on X_p for the canonical mixed finite-capacitary/Archimedean-direct- image metric.

13.1 The exact modular capacity pair at every prime

For $0 < Y < 1/p$, let the Archimedean pointed disc be

$$\iota_Y : \mathbb{D} \longrightarrow X_p(\mathbf{C}), \qquad q = e^{-2\pi Y} z, \tag{110}$$

and retain the integral Tate residue disc at every finite place $\ell \neq p$. At the distinguished place p, let $W_0(p)$ be Buzzard's maximal wide-open through the Tate component.

Lemma 13.10 (Rational skeleton exhaustion and exact finite capacity). *For every prime p there is an increasing sequence of connected wide-opens in $X^{\mathrm{an}}_{p,\mathbf{C}_p}$,*

$$\Omega_{p,1} \Subset \Omega_{p,2} \Subset \cdots \Subset W_0(p), \qquad \bigcup_{\nu\geq 1} \Omega_{p,\nu} = W_0(p),$$

such that every $\Omega_{p,\nu}$ descends to a finite extension of $\mathbf{Q}_p$, has affinoid complement, and is a rational wide-open: after that finite extension there is a rational function $u_{p,\nu}$, with polar divisor a positive multiple of P, for which

$$\Omega_{p,\nu} = \{|u_{p,\nu}|_p > 1\}.$$

If

$$\lambda_{p,\nu} = -\log \frac{\|\partial_q\|^{\mathrm{cap}}_{p,\Omega_{p,\nu}}}{\|\partial_q\|_{p,\mathrm{int}}}, \tag{111}$$

then

$$\lambda_{p,\nu} \longrightarrow L_p = \frac{12\log p}{p-1}. \tag{112}$$

Moreover $\Omega_{p,\nu+1}$ is a larger continuation neighborhood of the affinoid closure of $\Omega_{p,\nu}$, so every sufficiently large member, together with any fixed Archimedean q-disc of positive total tangent degree, is admissible for Theorem 13.7.

For $p \geq 5$, after a finite unramified extension splitting the supersingular residue points, put

$$\vartheta_\nu = \frac{\nu}{\nu+1}.$$

The domain $\Omega_{p,\nu}$ may be chosen by retaining, in the branch indexed by a geometric supersingular elliptic curve $E/\overline{\mathbf{F}}_p$, the initial skeleton segment of length $\vartheta_\nu i(E)$, where

$$i(E) = \frac{\#\,\mathrm{Aut}(E)}{2}.$$

Its effective resistance and tangent gain are then

$$R_{p,\nu} = \left(\sum_{[E]\ \mathrm{ss}} \frac{1}{\vartheta_\nu i(E)}\right)^{-1} = \vartheta_\nu \frac{12}{p-1}, \qquad \lambda_{p,\nu} = R_{p,\nu} \log p. \tag{113}$$

For $p = 2, 3$, one may instead take

$$\Omega_{p,\nu} = \{|f_p|_p < p^{\vartheta_\nu 12/(p-1)}\}, \qquad \lambda_{p,\nu} = \vartheta_\nu \frac{12 \log p}{p-1}. \tag{114}$$

All normalized local and global Arakelov degrees are independent of the finite extensions used to define these cuts.

Proof. Assume first that $p \geq 5$. Pass to a finite unramified extension over which all supersingular residue points split. By [13, §2, citing Deligne–Rapoport VI.6.16], the regular semistable skeleton between the two principal ordinary components consists of parallel paths indexed by the supersingular elliptic curves and having lengths $i(E)$. After a further finite ramified extension whose ramification index is divisible by $\nu + 1$, the point at distance $\vartheta_\nu i(E)$ on every path is a vertex of a semistable subdivision. Retain the component through P and the initial path segments up to those vertices. The tube of this connected subgraph is $\Omega_{p,\nu}$; its complement is a connected affinoid neighborhood of the opposite ordinary component. The inclusions are compact, and the retained lengths tend to the full branch lengths, so these domains increase to $W_0(p)$. This agrees with the affinoid exhaustion of the maximal wide-open on a fine tame cover in [5, Proposition 4.1].

The domain $\Omega_{p,\nu}$ is connected and contains P, while its complement $U_{p,\nu}$ is affinoid. Therefore [3, Proposition 5.6], applied with $D = P$, produces $u_{p,\nu}$ with polar divisor a positive multiple of P and $U_{p,\nu} = \{|u_{p,\nu}|_p \leq 1\}$. This is precisely the asserted rational lemniscate description. Notice that Proposition 5.6 is being applied separately to each fixed domain; no continuity assertion is attributed to it.

On the subdivided reduction graph, the vertical correction of [3, Proposition 5.1] is the voltage generated by one unit of current inserted at the vertex through P and grounded at the cut vertices. Thus the coefficient at the source vertex is the effective resistance of the parallel paths of lengths $\vartheta_\nu i(E)$. By the Eichler–Deuring mass formula [14, §5,§10],

$$\sum_{[E]\ \mathrm{ss}} \frac{1}{i(E)} = 2 \sum_{[E]\ \mathrm{ss}} \frac{1}{\#\,\mathrm{Aut}(E)} = \frac{p-1}{12}.$$

This proves the resistance formula in (113). If the ramification index used for the subdivision is e, a retained branch of normalized length $\vartheta_\nu i(E)$ consists of $e\vartheta_\nu i(E)$ unit model edges, while each such edge contributes $\log p/e$ to the normalized local degree. Adjunction, as in (85), therefore gives $\lambda_{p,\nu} = R_{p,\nu} \log p$, independently of the chosen ramified extension, and proves the asserted limit.

For $p = 2, 3$, the explicit calculations in [6, Lemma 3.1 and the proof of Theorem 3.5] give

$$W_0(p) = \{|f_p|_p < p^{12/(p-1)}\}.$$

The discs in (114) increase compactly to this domain. Their complements are affinoid, and [3, Corollary 5.8], together with $f_p = q + O(q^2)$ and adjunction, gives the tangent gain displayed there.

For every p, the next member of the sequence supplies the larger compact continuation neighborhood required in the finite-order estimates. Finally, the extensions above may depend on ν. Each prescribed finite local extension can be induced by a finite number-field base change; then Lemma A.2 shows that, after division by the extension degree, local degrees, determinant degrees, intersection numbers, and slopes inequalities are unchanged. This proves the lemma. $\square$

Theorem 13.11 (Exact modular capacity pair at every prime)**.** *For every prime p and $0 < Y < 1/p$, the exhaustion of Lemma 13.10, with normalized invariants after the harmless finite base changes used there, gives rational wide-opens*

$$\Omega_{p,1} \Subset \Omega_{p,2} \Subset \cdots \Subset W_0(p)$$

whose associated exact mixed continuation data α_ν satisfy

$$\lim_{\nu\to\infty} \Lambda_{\alpha_\nu} = L_p - 2\pi Y, \tag{115}$$

$$\lim_{\nu\to\infty} \Theta_{\alpha_\nu} = L_p - 2\pi Y, \tag{116}$$

$$\lim_{\nu\to\infty} \mathscr{H}_{\alpha_\nu} = L_p - 2\pi Y + C_p(Y), \tag{117}$$

where

$$L_p = \frac{12\log p}{p-1}$$

and $C_p(Y)$ is defined by (67). Equivalently, the maximal modular capacitary pair is

$$\boxed{(\Lambda_p(Y), \mathscr{H}_p(Y)) = \left(\frac{12\log p}{p-1} - 2\pi Y, \frac{12\log p}{p-1} - 2\pi Y + C_p(Y)\right).} \tag{118}$$

For every jointly rational packet $\mathcal{S}$ and every $1 < \xi < d_p m$,

$$\boxed{\tau^{\mathrm{curve}}_{p,\mathcal{S}}(\xi) \geq \left(1 - \frac{1}{m}\right)\left(\frac{12\log p}{p-1} - 2\pi Y\right) - \frac{C_p(Y)}{m}.} \tag{119}$$

Proof. By Lemma 13.10, the distinguished finite-place contribution of the rational inner domains tends to

$$L_p = \frac{12\log p}{p-1}.$$

At every finite place $\ell \neq p$ the chosen domain is the integral Tate residue disc, so [3, Example 5.11] contributes zero relative tangent degree. At infinity, (110) contributes $-2\pi Y$. This proves (115).

Target normal and Archimedean overflow. No interior point of the upper half-plane represents the cusp P. Hence

$$\iota_Y^* P = o$$

on the pointed disc. The exact normal decomposition (138) therefore has no additional target-normal collision, proving (116).

It remains to evaluate the Archimedean term $\mathscr{W}$. On the boundary write $\tau = x + iY$, $0 \leq x < 1$. Nonidentity self-collisions of the modular q-disc are parametrized by primitive bottom rows (c, d) with $c > 0$ and $p \mid c$. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$,

$$\Im(\gamma\tau) = \frac{Y}{(cx+d)^2 + c^2Y^2}.$$

The source Green function is $2\pi(\Im\tau - Y)$, so the direct-image Green–Jensen formula gives

$$\mathscr{W}_p(Y) = \sum_{\substack{c>0 \\ p|c}} \sum_{\substack{d\in\mathbf{Z} \\ (c,d)=1}} \int_0^1 2\pi Y \left(\frac{1}{(cx+d)^2 + c^2Y^2} - 1\right)_+ \mathrm{d}x. \tag{120}$$

Elliptic stabilizers occur only over a measure-zero set and do not alter this integral. For fixed c, the summand vanishes when $cY \geq 1$. When $cY < 1$, unfolding the primitive residues $d \bmod c$ and putting $b_c = \sqrt{1 - c^2Y^2}$ gives

$$\begin{aligned} &\frac{2\pi Y\varphi(c)}{c} \int_{-b_c}^{b_c} \left(\frac{1}{u^2 + c^2Y^2} - 1\right) \mathrm{d}u \\ &\qquad = \frac{4\pi\varphi(c)}{c^2} \left(\arccos(cY) - cY\sqrt{1 - c^2Y^2}\right). \end{aligned}$$

Thus

$$\mathscr{W}_p(Y) = C_p(Y). \tag{121}$$

Together with (139), this proves (117).

The rational skeleton exhaustion of Lemma 13.10 supplies the required inner data. Use the same Archimedean disc for every member. Its distinguished finite contribution tends to L_p, while the target-normal comparison has no extra cusp preimage and the Archimedean term is the fixed number $C_p(Y)$. Consequently the exact mixed invariants converge to (115)–(117). Since $\log p > 1 - 1/p$, one has $L_p > 12/p > 2\pi/p > 2\pi Y$; hence every sufficiently large member is pseudoconcave. Apply Theorem 13.7 to those members and pass to the limit. The arithmetic type is independent of the continuation datum, so this yields (119). □

Remark 13.12 (Why genus zero was inessential for the collision formula). The proof of (121) uses only the uniformizing quotient $\Gamma_0(p)\backslash\mathfrak{H}$, the width-one cusp, and the source Green function. A Hauptmodul is unnecessary. Consequently the formula $\mathscr{H}_p(Y) = \Lambda_p(Y) + C_p(Y)$ from Proposition 8.3 is intrinsic and extends to every prime level.

14 All-prime density one

Proof of Theorem 1.6. Fix $\beta > 1$. Let $\mathcal{S}_j$ be the rational odd arguments in the shell

$$A = \beta^j \leq s < \beta A,$$

and put $k_j = \#\mathcal{S}_j$. Empty shells may be ignored. For a nonempty shell,

$$k_jA \leq m - 1 < k_j\beta A, \qquad s_* < \beta A, \qquad Q_{\mathcal{S}_j} < \beta A m.$$

Choose $\xi = 1 + 1/\log A$. Since d_p is fixed,

$$G_{d_p,m}(\xi) = d_p \log m + O_p(1),$$

and (98) gives

$$\tau^{\mathrm{curve}}_{p,\mathcal{S}_j}(\xi) \leq \frac{2d_p\beta}{k_j} \log m + O_{p,\beta}\left(\frac{1}{k_j}\right). \tag{122}$$

In Theorem 13.11 choose $Y = 1/m$; for a nonempty shell this is $< 1/p$ once j is sufficiently large. Since

$$0 \leq \arccos x - x\sqrt{1 - x^2} \leq \frac{\pi}{2}, \qquad \varphi(c) \leq c,$$

we have

$$C_p(1/m) \le \frac{2\pi^2}{p} H_{\lfloor (m-1)/p \rfloor} = O_p(\log m). \tag{123}$$

The explicit exact inequality (119) therefore gives

$$\tau^{\mathrm{curve}}_{p,\mathcal{S}_j}(\xi) \ge L_p - O_p\left(\frac{\log m}{m}\right), \qquad L_p = \frac{12\log p}{p-1}. \tag{124}$$

Initially $k_j = O_\beta(A)$ and hence $m = O_\beta(A^2)$, so $\log m = O_\beta(\log A)$. The right side of (124) is at least $L_p/2$ for all sufficiently large j. Combining the upper and lower estimates first gives

$$k_j = O_{p,\beta}(\log A).$$

It follows that

$$m = O_{p,\beta}(A \log A), \qquad \log m = \log A + O_{p,\beta}(\log\log A).$$

Substitution back yields

$$k_j \le \left(\frac{2d_p\beta}{L_p} + o_p(1)\right) \log A. \tag{125}$$

If $\beta^J \le X < \beta^{J+1}$, summing over $j \le J$ gives

$$R_p(X) \le \left(\frac{d_p\beta}{L_p \log\beta} + o_p(1)\right) (\log X)^2. \tag{126}$$

The function $\beta/\log\beta$ is minimized at $\beta = e$, so

$$\frac{ed_p}{L_p} = \frac{e\,d_p(p-1)}{12\log p},$$

which proves (15). When $g_p \ge 1$, substitute $d_p \le 2g_p$ from Theorem 11.2 to obtain (16).

For each shell, the packet is fixed before the source-degree limit $D \to \infty$ is taken. Packet-dependent module generators, bad primes, metric comparisons, and boundary norms have therefore disappeared into $o(D^2)$ before $j \to \infty$. The subsequent limits in the capacitary exhaustion and in j introduce no packet-uniform auxiliary-prime assumption. □

Remark 14.1 (What the all-prime theorem does not assert). The coefficient in (15) is explicit for any chosen finite map of degree d_p, but the general construction $d_p \le 2g_p$ need not be gonality-optimal. Improving the map can therefore improve the displayed constant and the finite-packet bounds below. No uniformity as $p \to \infty$ beyond the stated formula is claimed, and the theorem still permits an infinite polylogarithmically sparse set of rational exceptions.

14.1 Certified finite packets at positive genus

The exact capacity formula permits a finite-packet certificate that does not enumerate subsets.

Proposition 14.2 (One-dimensional finite-packet certificate)**.** *Fix a prime p, let d_p be an admissible finite-map degree in Theorem 11.2, and let $\bar d \ge d_p$ be a certified upper bound. Fix an odd cutoff S and an integer $k \ge 1$. Put*

$$m_0(k,x) = x + k^2, \tag{127}$$

$$U_{\bar d,k}(x) = 2\bar d\left[\frac{x - 2 + x\log m_0(k,x)}{m_0(k,x)} + \frac{x+2}{m_0(k,x)^2}\right], \tag{128}$$

$$B_{p,k}(x) = \left(1 - \frac{1}{m_0(k,x)}\right)\left(L_p - \frac{2\pi}{m_0(k,x)}\right) - \frac{2\pi^2}{p\,m_0(k,x)} H_{\lfloor (m_0(k,x)-1)/p \rfloor}. \tag{129}$$

Assume, for every relevant x, that

$$m_0(k,x) > \max\{p, \bar{d}\}, \qquad L_p - \frac{2\pi}{m_0(k,x)} > 0.$$

If

$$B_{p,k}(x) > U_{\bar{d},k}(x) \qquad (2k+1 \le x \le S,\ x \text{ odd}), \tag{130}$$

then among the odd arguments $3, 5, \ldots, S$ at most $k-1$ values $\zeta_p(s)$ are rational.

Proof. Suppose instead that k distinct values are rational and let $\mathcal{S}$ be one such k-element packet, with $x = \max \mathcal{S}$. The other $k-1$ arguments are distinct positive odd integers at least 3, so their sum is at least

$$3 + 5 + \cdots + (2k-1) = k^2 - 1.$$

Thus

$$m = 1 + \sum_{s \in \mathcal{S}} s \ge x + k^2 = m_0(k,x). \tag{131}$$

For every $s \in \mathcal{S} \setminus \{x\}$ one has $s \le x-2$, and hence $s^2 \le (x-2)s$. Therefore

$$Q_{\mathcal{S}} \le x^2 + (x-2)(m-1-x) = (x-2)m + x + 2. \tag{132}$$

Choose $\xi = \bar{d}$. The assumption $\bar{d} < m_0(k,x) \le m$ makes this cutoff admissible. Since $\bar{d} \ge d_p$, the second branch of (82) gives

$$G_{d_p,m}(\bar{d}) = d_p \log\left(\frac{d_p m}{\bar{d}}\right) \le d_p \log m \le \bar{d} \log m.$$

Equations (98) and (132) therefore give

$$\tau^{\text{curve}}_{p,\mathcal{S}}(\bar{d}) \le 2\bar{d}\left[\frac{x - 2 + x\log m}{m} + \frac{x+2}{m^2}\right].$$

The derivative of the quantity in square brackets is

$$\frac{2 - x\log m}{m^2} - \frac{2(x+2)}{m^3} < 0$$

for $x \ge 3$ and $m \ge 3$. Hence (131) implies

$$\tau^{\text{curve}}_{p,\mathcal{S}}(\bar{d}) \le U_{\bar{d},k}(x). \tag{133}$$

On the analytic side use Theorem 13.11 with the fixed choice $Y = 1/m_0(k,x)$. Write

$$A = L_p - \frac{2\pi}{m_0(k,x)}.$$

For fixed Y, the exact lower side is

$$\left(1 - \frac{1}{m}\right) A - \frac{C_p(Y)}{m} = A - \frac{A + C_p(Y)}{m},$$

which is increasing in m because $A > 0$ and $C_p(Y) \ge 0$. Moreover (123), with m_0 in place of m, gives

$$C_p(1/m_0) \le \frac{2\pi^2}{p} H_{\lfloor (m_0-1)/p \rfloor}.$$

Consequently joint rationality forces

$$\tau^{\text{curve}}_{p,\mathcal{S}}(\bar{d}) \ge B_{p,k}(x). \tag{134}$$

Equations (133) and (134) contradict (130). □

Proof of Theorem 1.4. Apply Proposition 14.2 with $S = 2401$, $\bar{d} = 2g_p$, and the values shown below. The certificate checks every admissible odd largest argument x, using outward-rounded integer fixed-point intervals at scale 10^{45}. The harmonic numbers in (129) are enclosed by summing the individual rational terms outward; π is enclosed by Machin's formula and alternating arctangent series, and logarithms are enclosed after dyadic range reduction by the positive arctanh series. No binary floating-point value enters a certificate decision.

p	$\bar{d}$	excluded k	rational cap	worst x	certified margin
11	2	191	190	2401	> 0.019624155425482
17	2	227	226	2401	> 0.006340667081541
19	2	238	237	2401	> 0.013201735527294
23	4	378	377	2401	> 0.006562285899278
29	4	415	414	2401	> 0.005457529203788
31	4	426	425	2401	> 0.002866264645707
37	4	458	457	2401	> 0.002720591459553
41	6	596	595	2401	> 0.003315013860040
43	6	607	606	2401	> 0.000684777792175

Subtracting the rational caps from 1200 gives the irrational counts in Theorem 1.4.

For the final column of that theorem, if a prefix contains N odd arguments, we apply the same certificate with $k = N - 999$: excluding such a packet leaves at most $N - 1000$ rational values. The certified cutoffs are respectively

$$2379,\ 2459,\ 2483,\ 2823,\ 2921,\ 2951,\ 3039,\ 3445,\ 3483.$$

The stored certificate also checks the immediately preceding cutoff for this sufficient criterion and records a nonpositive lower margin there. The certificate files are listed in Section 9.3. □

15 Logical dependency and adversarial audit

Gate	Input	Use and final audit conclusion
Single blocks	Calegari; BGG/Bol; internal Section 3	Constructs each algebraic extension and exact raw rows for general p.
Joint rank and descent	Internal Theorem 4.1 and Lemma 11.1	Even symmetric powers descend away from the fixed elliptic divisor; the multiplicity-free direct sum has full monodromy rank.
Genus-zero zero estimate	CDT Theorem 3.2.13; Corollary 4.3	Applies on $\mathbf{P}^1$ and retains the sharp normalized interval $m + o(1)$.
General-curve zero estimate	CDT Theorem 3.2.8; Theorem 11.2	Restriction of scalars gives a derivative-stable rank-$d_p m$ system; d_p enlarges only the pivot range, not the genuine source rank.
Hasse saving	Katz; Proposition 5.1 and Theorem 5.3	Used only for the sharp genus-zero theorems; not load-bearing for arbitrary-p density one.
Genus-zero Cartier	Exact Dwork; Lemma 6.1 and Proposition 6.2	Common Hauptmodul Frobenius and full-product digitization give the CDT degree-D window.
General-curve Cartier	Exact q-Dwork; Lemmas 12.1 and 12.2 and Proposition 12.3	Global divisor control makes the short q-jet injective; exact $q \mapsto q^{\ell}$ removes all Taylor correction terms.

Fitting polygon	Internal Propositions 6.4 and 6.6	Counts ℓ-adic thickness $s-b+1$; used for strengthening, not for the certified finite counts.
Genus-zero Bost bridge	Bost, Bost–Charles, CDT; Lemma 7.1	The $\mathbf{P}^1$ fixed-bundle comparison changes only $o(D^2)$.
General-curve fixed-twist bound	Finite generation; Bost–Chambert-Loir Theorem 6.1, (6.7)–(6.8); Lemma 13.3	A fixed packet is a quotient of finitely many fixed scalar twists. Relative Serre vanishing identifies the quotient lattices with the good-prime coefficient lattices. Packet dependence is $O(\log D)$ per pivot, while the coefficient of D depends only on p.
Scalar capacitary jets	Bost–Chambert-Loir Proposition 5.15; Cauchy; Lemma 13.4	Contraction is performed first, leaving a scalar section. The arbitrary reference tangent norm cancels formally and gives the term $-n\Lambda_p^\circ$.
Hybrid compatibility	Raw depth; intrinsic Cartier; Proposition 13.5	Shrinking the finite source lattice improves analytic operator norms. Small-prime residuals require proper prime powers; coherent models eliminate any hidden packet-dependent D^2 term at fixed bad primes. Large primes give the stated window integral.
$p = 13$ continuation	Maier; Deuring; Buzzard; Lemma 8.1	Identifies $W_0(13) = \{\lvert f_{13}\rvert_{13} < 13\}$ and permits explicit finite estimates.
All-prime positive width	Buzzard; Bost–Chambert-Loir Proposition 5.1, the proof of Proposition 5.6, Example 5.11, and Proposition 5.15; Lemma 13.1 and Proposition 13.2	The capacitary vertical coefficient on the component met by the cusp is strictly positive, so adjunction makes the local tangent norm strictly smaller than the integral-disc norm; a complex q-disc consumes arbitrarily little of that gain.
Exact mixed capacitary normalization	Bost–Chambert-Loir Propositions 5.1, 5.6, and 5.17, and Corollary 5.20; Bost–Charles Lemma 7.2.7, Proposition 7.2.10, and the meromorphicity criterion of §7.3; Chen–Moriwaki Corollary 4.5.2; Theorem 13.7 and Section A	The finite capacitary $\mathbf{Q}$-model and Archimedean direct-image Green functions define one nef and big mixed divisor. Denominator clearing realizes the capacitary tangent line as a pseudoconcave normal line, but neither a whole-divisor direct image nor a meromorphic map is inferred from that normal-line isometry. The exact source formula uses the Chen–Moriwaki determinant theorem; Bost–Charles Theorem 8.2.5 is a theta formula and is not used as a determinant formula. A nested lemniscate gives the uniform finite-order estimate and the exact target-normal Θ-cancellation.

Exact modular capacity pair	Buzzard Proposition 4.1; Coleman–McMurdy; Eichler–Deuring mass formula; Bost–Chambert-Loir Propositions 5.1 and 5.6; Lemma 13.10 and Theorem 13.11	After an unramified splitting extension, the supersingular annuli for $p \geq 5$ are parallel resistors of widths $\#\,\mathrm{Aut}(E)/2$. Rational skeleton cuts give inner wide-opens whose effective resistances converge to $12/(p-1)$; Proposition 5.6 is used only to realize each cut-domain as a rational lemniscate. The wild primes $2, 3$ use their explicit Hauptmodul discs. The Archimedean collision sum is intrinsic to the $\Gamma_0(p)$ quotient and is genus-independent.
Quantitative genus-zero density	Internal Section 10	Uses exact modular width and collision formula, yielding the coefficient $e(p-1)/(12\log p)$.
All-prime density	Internal Theorem 13.11 and Section 14	Uses the exact capacity pair with $Y = 1/m$ and multiplicative shells, yielding the explicit coefficient $ed_p(p-1)/(12\log p)$. The robust theorem remains an independent qualitative proof.
Numerics	Independent fixed-point certificates; Propositions 9.2 and 14.2	The genus-zero counts retain their original certificates. The new positive-genus table uses the certified upper bound $\bar{d} = 2g_p$, checks every possible largest packet argument with outward-rounded integer fixed-point arithmetic, and uses no binary floating point.

Warning 15.1 (Distinct weights). The functional-rank proof uses that the representations $\mathrm{Sym}^{s-1}(\mathbf{C}^2)$ occur with multiplicity one. Repeated copies of the same weight can support diagonal subrepresentations and do not automatically add functional rank.

Warning 15.2 (No scalarization). The proof does not construct global scalar coordinates carrying literal depths $0, 1, \ldots, s$. The small-prime saving uses only the mod-ℓ Hasse comparison of the genuine deepest multiplier. The large-prime argument stays on the de Rham frame torsor and digitizes the complete truncated products.

Warning 15.3 (Counting versus individual identification). Theorems 1.3 and 1.4 bound the number of rational values in finite ranges but does not identify every possible exception. Proving long explicit lists of individual irrational values requires a different amplification: an unconditional arithmetic anchor packet that contributes functional rank under the rationality assumption for one target only.

Warning 15.4 (All-prime scope). The all-prime theorem is a fixed-p assertion. Its leading density coefficient is now explicit once a finite map degree d_p is chosen, and the safe construction $d_p \leq 2g_p$ is uniform at fixed level. The coefficient is not claimed optimal: the displayed choice $2g_p$ is only an upper bound for the actual degree, and a lower-degree map unramified at the Tate cusp improves both the density constant and the finite-packet arithmetic cost. The sharp Hasse-improved type and denominator polygon remain genus-zero results, whereas the exact modular capacity and collision formulas are valid at every prime.

Warning 15.5 (Exact-normalization scope). The normalization-robust theorem still supplies an independent qualitative proof of density one, but the explicit leading coefficient in Theorem 1.6 uses the exact mixed normalization and the capacity exhaustion of Theorem 13.11. The exact result Theorem 13.7 realizes the metrized tangent line, not the entire finite vertical

Bost–Chambert-Loir divisor, as the normal line of a formal-analytic surface. It also does not infer a Bost–Charles meromorphic map from an arbitrary identification of generic formal discs. The finite divisor is used directly in the mixed canonical metric, and the determinant source formula is supplied by Chen–Moriwaki rather than by the Bost–Charles theta Hilbert–Samuel theorem. These distinctions are recorded explicitly in Warning A.5.

A Exact mixed capacitary normalization and the normal-line bridge

This appendix proves Theorem 13.7. The construction is adelic but deliberately asymmetric. At finite places we use the Bost–Chambert-Loir capacitary $\mathbf{Q}$-model itself. At Archimedean places we use the direct-image Green function of the chosen continuation map in the sense of Bost–Charles. The first audit point is that these two pieces define one semipositive adelic divisor on the modular curve. The second is that the capacitary tangent line, after denominator clearing, is a genuine normal line of a pseudoconcave formal-analytic surface.

No compatible Bost–Charles meromorphic map from that formal-analytic surface to the modular curve is asserted or needed. Such a map is a separate fraction-field condition in the formal category and does not follow merely from an identification of generic formal discs. Likewise, an isometry of normal lines does not identify the full finite vertical divisor. The exact source and jet formulas below are therefore proved directly on X_p, rather than by invoking an unproved whole-divisor direct-image realization.

A.1 The canonical mixed capacitary divisor

Fix p and the continuation datum

$$\boldsymbol{\alpha} = ((\Omega_v, o_v, \iota_v))_v = \boldsymbol{\Omega}_p^{\circ}$$

of Proposition 13.2. At a finite place, ι_v is the inclusion of a connected wide-open neighborhood of P in X_p^{an}; at all but finitely many places it is the integral Tate residue disc. At an Archimedean place σ, write

$$g_\sigma = g_{\Omega_\sigma, o_\sigma}, \qquad \mu_\sigma = \mu_{\Omega_\sigma, o_\sigma}$$

for the equilibrium Green function and harmonic measure. The map ι_σ is nonconstant and unramified at o_σ.

At the finite places, [3, Propositions 5.1 and 5.17] supplies a normal flat projective model $\mathscr{X}_\alpha$ of X_p and a vertical $\mathbf{Q}$-divisor V_α^{fin} such that

$$\mathscr{P}_\alpha^{\mathrm{fin}} = \mathscr{P} + V_\alpha^{\mathrm{fin}}$$

induces the capacitary metric of Ω_v on $\mathcal{O}(P)$ at every finite place. After one fixed birational refinement, which does not change these metrics or their intersection numbers, we may and do assume that the section P lies in the regular locus of $\mathscr{X}_\alpha$; thus adjunction and $P^*\mathcal{O}(P)$ are literal near the section. The adelic tube is defined using the integral Tate cusp model at every finite place other than p and an adapted model of the chosen rational wide-open at p. Proposition 5.17, through its Raynaud comparison and Moret–Bailly descent step, places these local data on one global model. The corresponding vertical correction is zero at every integral Tate place. In Proposition A.12 the coefficient-bundle models used in Section 12 are chosen from this same model after one common refinement and enlargement of the fixed exceptional set. The finite vertical divisor is independent of the auxiliary Archimedean components by the local uniqueness in Proposition 5.1.

For an Archimedean embedding σ, let

$$g_{\alpha,\sigma} = \iota_{\sigma *} g_\sigma \tag{135}$$

be the direct-image Green function for P in the sense of [4, Sections 5.1 and 7.2]. It has logarithmic singularity of multiplicity one at P, because ι_σ is unramified at o_σ. Its curvature measure is the positive direct image of the source equilibrium measure.

Definition A.1 (Canonical mixed divisor)**.** Define

$$\widehat{P}_\alpha = \left(\mathscr{P}_\alpha^{\mathrm{fin}}, (g_{\alpha,\sigma})_\sigma\right) \in \widehat{\mathrm{Div}}(\mathscr{X}_\alpha)_{\mathbf{Q}} \tag{136}$$

and put

$$\overline{L}_\alpha = \mathcal{O}(\widehat{P}_\alpha).$$

Let $\overline{T}_\alpha$ denote the line $T_P X_p$ with the finite Bost–Chambert-Loir capacitary metrics and, at infinity, the source capacitary metric on $T_{o_\sigma}\Omega_\sigma$ transported through $D\iota_\sigma(o_\sigma)$. Finally put

$$\Lambda_\alpha = \widehat{\deg}\overline{T}_\alpha, \qquad \Theta_\alpha = \widehat{\deg} P^*\overline{L}_\alpha, \qquad \mathscr{H}_\alpha = \widehat{P}_\alpha^2. \tag{137}$$

The distinction between Λ_α and Θ_α is essential. The former is a source-tangent degree. The latter is the degree of the target normal line determined by the direct-image Green function. Other preimages of P under ι_σ may make the two different.

Lemma A.2 (Base-change and denominator invariance)**.** *Let $K/\mathbf{Q}$ be a finite extension. Pull back all models, $\mathbf{Q}$-divisors, metrized lines, sources, and jet targets to $\mathcal{O}_K$. Then every Arakelov degree and arithmetic intersection number is multiplied by $[K:\mathbf{Q}]$. Consequently all identities and inequalities in this appendix are unchanged after division by $[K:\mathbf{Q}]$. The same conclusions hold for an Arakelov $\mathbf{Q}$-divisor after multiplying by a common denominator and dividing the resulting degree or intersection number by the corresponding first or second power.*

Proof. The local degree at a place v is repeated with the usual local-degree weights over the places of K above v; their sum is $[K:\mathbf{Q}]$. This proves the assertion for metrized lines and for determinant degrees. The projection formula gives the same scaling for arithmetic intersections. Homogeneity in a common denominator gives the final assertion. $\square$

Proposition A.3 (Nefness, bigness, and the exact normal decomposition)**.** *The divisor $\widehat{P}_\alpha$ is effective, nef, and big. Moreover*

$$\Theta_\alpha - \Lambda_\alpha = \sum_\sigma \int_{\Omega_\sigma} g_\sigma \, \delta_{\iota_\sigma^* P - o_\sigma} \geq 0 \tag{138}$$

and

$$\boxed{\mathscr{H}_\alpha = \Theta_\alpha + \mathscr{W}_\alpha, \qquad \mathscr{W}_\alpha = \sum_\sigma \int_{X_{p,\sigma}(\mathbf{C})} g_{\alpha,\sigma}\, c_1(\overline{L}_{\alpha,\sigma}) = \sum_\sigma \int_{\Omega_\sigma} g_\sigma \, \iota_\sigma^* c_1(\overline{L}_{\alpha,\sigma}) \geq 0.} \tag{139}$$

In particular,

$$\Theta_\alpha \geq \Lambda_\alpha > 0, \qquad \mathscr{H}_\alpha > 0.$$

The finite capacitary corrections occur in Θ_α exactly once; the term $\mathscr{W}_\alpha$ is purely Archimedean.

Proof. At each finite place, the vertical correction in $\mathscr{P}_\alpha^{\mathrm{fin}}$ is the solution of the graph Dirichlet problem of [3, Proposition 5.1]. The resulting $\mathbf{Q}$-divisor is effective. It has intersection zero with every vertical component occurring in its support and nonnegative intersection with every other vertical component. At infinity, the direct-image Green function is nonnegative and its curvature is the positive direct image of the source equilibrium measure. Hence $\widehat{P}_\alpha$ is effective and defines a semipositive adelic metric on $\mathcal{O}(P)$.

At a finite place the continuation map is the inclusion, so the source and target capacitary tangent metrics agree. At an Archimedean place, the local normal comparison in the proof of [4, Lemma 7.2.7 and Proposition 7.2.10], with ramification index one, gives

$$\widehat{\deg}_\sigma P^*\overline{L}_\alpha - \widehat{\deg}_\sigma \overline{T}_\alpha = \int_{\Omega_\sigma} g_\sigma \, \delta_{\iota_\sigma^* P - o_\sigma}.$$

The divisor $\iota_\sigma^* P - o_\sigma$ is effective. Summing proves (138), so the horizontal component P has strictly positive arithmetic degree. The numerical criterion of [1, Proposition 6.9], used in [3, Proposition 7.7], now proves nefness. Indeed, the curvature is nonnegative, the degree on P is positive, the degree on every vertical component in the support is zero, and the remaining vertical degrees are nonnegative; every horizontal curve different from P has nonnegative degree because both the finite divisor and the Green functions are effective.

Write $\mathscr{P}_\alpha^{\mathrm{fin}} = \mathscr{P} + V$. By the orthogonality in [3, Proposition 5.1], $\mathscr{P}_\alpha^{\mathrm{fin}} \cdot V = 0$. The induction formula for the arithmetic intersection pairing, in the normalization used in the proof of [4, Proposition 7.2.6], therefore gives

$$\widehat{P}_\alpha^2 = \widehat{\deg} P^* \overline{L}_\alpha + \sum_\sigma \int_{X_{p,\sigma}(\mathbf{C})} g_{\alpha,\sigma} \, c_1(\overline{L}_{\alpha,\sigma}).$$

The projection formula for direct-image Green functions gives the second expression in (139). Both factors in its integrand are nonnegative. Thus the self-intersection is positive. We have already proved that $\overline{L}_\alpha$ is nef; a nef Arakelov divisor of positive self-intersection is nef and big in the sense of [4, Section 8.2.2]. This proves the proposition. $\square$

A.2 The capacitary normal-line realization

The preceding proposition already constructs the exact target metric needed for arithmetic Hilbert–Samuel. The following result closes the separate formal-analytic realization issue. Its statement is intentionally about the normal line, not the entire finite vertical divisor.

Proposition A.4 (Denominator-cleared capacitary normal realization)**.** *After a finite extension $K/\mathbf{Q}$ clearing the rational finite-place exponents of $\overline{T}_\alpha$, there exists a smooth formal-analytic arithmetic surface*

$$\widetilde{\mathscr{V}}_\alpha = \left(\widehat{\mathscr{V}}_\alpha, (\Omega_\sigma, o_\sigma, \gamma_\sigma)_\sigma\right) \quad \textit{over } \mathcal{O}_K$$

whose metrized normal line along its canonical section is isometric to $\overline{T}_\alpha \otimes_{\mathbf{Q}} K$. Consequently

$$\frac{1}{[K:\mathbf{Q}]} \widehat{\deg} N_{P_K} \widetilde{\mathscr{V}}_\alpha = \Lambda_\alpha > 0, \tag{140}$$

so $\widetilde{\mathscr{V}}_\alpha$ is pseudoconcave.

Proof. By [3, Propositions 5.1 and 5.17], the finite metrics on $T_P X_p$ are $\mathbf{Q}$-model metrics and are integral at all but finitely many places. Choose a common denominator for their valuation exponents and a finite extension $K/\mathbf{Q}$ whose ramification indices above the exceptional places are divisible by that denominator; one may take a compositum of suitable Eisenstein extensions at the finitely many rational primes involved. The unit balls of the pulled-back metrics are then honest rank-one $\mathcal{O}_{K,w}$-lattices. Since only finitely many differ from the standard tangent lattice, they are the localizations of one invertible fractional $\mathcal{O}_K$-module $\mathcal{N}_\alpha \subset T_P X_p \otimes_{\mathbf{Q}} K$.

Set

$$\widehat{\mathscr{V}}_\alpha = \operatorname{Spf} \widehat{\operatorname{Sym}}_{\mathcal{O}_K}(\mathcal{N}_\alpha^\vee).$$

This is a smooth one-dimensional formal scheme over $\mathcal{O}_K$, with zero section $P_K \simeq \operatorname{Spec} \mathcal{O}_K$ and normal module $\mathcal{N}_\alpha$. For each embedding $\sigma : K \hookrightarrow \mathbf{C}$, choose a formal coordinate at o_σ and a conjugation-compatible formal isomorphism

$$\gamma_\sigma : \widehat{\mathscr{V}}_{\alpha,\sigma} \xrightarrow{\sim} \widehat{\Omega}_{\sigma,o_\sigma}$$

whose differential identifies the complex fibre of $\mathcal{N}_\alpha$ with $T_{o_\sigma}\Omega_\sigma$ after transport through $D\iota_\sigma(o_\sigma)$. In dimension one the derivative may be prescribed by multiplying a local coordinate by a nonzero complex scalar; the choices may be paired under complex conjugation. Equip the complex fibre of $\mathcal{N}_\alpha$ with the corresponding source capacitary norm. The data then form a smooth formal-analytic arithmetic surface in the sense of [4, Sections 6.1–6.2], and its metrized normal line is precisely $\overline{T}_\alpha \otimes_{\mathbf{Q}} K$.

Taking normalized Arakelov degrees and applying Lemma A.2 gives (140). Positivity is Proposition 13.2; by definition, it is the pseudoconcavity of $\widetilde{\mathscr{V}}_\alpha$. □

Warning A.5 (Normal-line realization is not a map). An isometry of metrized normal lines determines neither a vertical divisor on the whole special fibre nor a Bost–Charles meromorphic map to $\mathscr{X}_{\alpha,K}$. In the formal category, meromorphicity requires the coordinate functions of the generic map to lie in the fraction field of the formal function ring and to become regular after a modification; an arbitrary identification of generic formal discs does not establish this condition. Accordingly, Proposition A.4 makes no such assertion. The exact normalization below uses the Bost–Chambert-Loir finite divisor itself and is proved directly on the modular curve.

A.3 Exact arithmetic Hilbert–Samuel for the source

Regard $\overline{L}_\alpha$ as a semipositive adelic line bundle on the smooth projective curve $X_p/\mathbf{Q}$. Its underlying line bundle $L = \mathcal{O}(P)$ has degree one and is therefore ample. Give the fixed bundle $\mathcal{F}_\mathcal{S}$ any fixed dominated adelic metric compatible with its integral models. On spaces of sections use the induced supremum norms; at every Archimedean place replace the supremum unit ball by its John ellipsoid in order to obtain a Hermitian norm.

Proposition A.6 (Exact determinant source formula)**.** *For every fixed packet $\mathcal{S}$,*

$$\widehat{\deg}\,\overline{M}_D^\alpha = \frac{m}{2}\mathscr{H}_\alpha D^2 + o_{p,\mathcal{S}}(D^2). \tag{141}$$

Replacing the supremum norms by the John norms changes the degree by $O_{p,\mathcal{S}}(D\log(D+2))$. Changing the fixed adelic metric or the fixed saturated integral model on $\mathcal{F}_\mathcal{S}$ changes it by $O_{p,\mathcal{S}}(D)$.

Proof. The curve X_p is geometrically integral and normal, the line $L = \mathcal{O}(P)$ is ample, and every local metric of $\overline{L}_\alpha$ is a dominated semipositive metric: at finite places it is a model metric, while at infinity it is the continuous Green metric of Proposition A.3. The vector-bundle arithmetic Hilbert–Samuel theorem [9, Corollary 4.5.2] therefore applies directly to the adelic vector bundle $\overline{\mathcal{F}}_\mathcal{S}$ and gives

$$\lim_{D\to\infty} \frac{\widehat{\deg}\,H^0(X_p, L^D \otimes \mathcal{F}_\mathcal{S})}{D^2/2} = \operatorname{rank}(\mathcal{F}_\mathcal{S})(\overline{L}_\alpha^2) = m\mathscr{H}_\alpha.$$

This is (141) for the adelic supremum norms.

If $r_D = \dim H^0(X_p, L^D \otimes \mathcal{F}_{\mathcal{S}})$, John's theorem for a symmetric convex body gives a determinant distortion at most $\frac{1}{2} r_D \log r_D$ at each Archimedean place. Since $r_D = mD + O(1)$, this is $O(D \log D)$. Two fixed dominated metrics on $\mathcal{F}_{\mathcal{S}}$ are uniformly equivalent place by place and agree integrally outside a fixed finite set; their determinant distortion is $O(r_D) = O(D)$. The same applies to two fixed saturated models of the coefficient bundle. This proves all assertions. □

Remark A.7 (Determinant versus theta asymptotics). The input used here is the determinant-degree formula of [9, Corollary 4.5.2]. The arithmetic Hilbert–Samuel formula in [4, Theorem 8.2.5] is instead a formula for a John-theta invariant. No unproved identification of that theta invariant with the determinant degree of the present source lattice is used.

A.4 Uniform finite-order jets and the Θ-cancellation

Let $\overline{C}_{\mathcal{S}}$ denote the fixed metrized line obtained from $\mathcal{F}_{\mathcal{S}}|_P$ and from the fixed contraction pairing. For a pivot n, the natural target is

$$\overline{J}^{\alpha}_{D,n} = P^* \overline{L}_{\alpha}^{D} \otimes \overline{T}_{\alpha}^{\vee \otimes n} \otimes \overline{C}_{\mathcal{S}}. \tag{142}$$

The source quotient is endowed with the quotient norm from $\overline{M}^{\alpha}_{D}$. Choose any nonzero rational vector in $J^{\alpha}_{D,n}$ and use it to identify the target with the standard scalar line. Denote the resulting local and global heights by $h^{\mathrm{sc}}_{v,\alpha}$ and h^{sc}_{α}. The product formula gives the exact, trivialization-independent identity

$$h^{\mathrm{sc}}_{\alpha}(\psi^{(n)}_D) = h^{\mathrm{nat}}_{\alpha}(\psi^{(n)}_D) + \widehat{\deg} \overline{J}^{\alpha}_{D,n}. \tag{143}$$

At the distinguished place p set $\delta_{p,D,n} = 0$; its analytic operator norm is controlled directly by the capacitary metric. At every finite place $v \neq p$, retain the coefficient excess $\delta_{v,D,n}$ defined in Section 13.

Lemma A.8 (Exact finite-place capacitary Schwarz estimate). *At the distinguished continuation place p, uniformly for all $D \geq 0$ and all $n \geq 0$,*

$$\log \|\psi^{(n)}_D\|_{p,\mathrm{nat}} \leq O_{p,\mathcal{S}}(1). \tag{144}$$

At every finite place $v \neq p$,

$$\log \|\psi^{(n)}_D\|_{v,\mathrm{nat}} \leq \delta_{v,D,n} + O_{v,p,\mathcal{S}}(1), \tag{145}$$

and outside a fixed finite set the error term is zero.

Proof. At p use the nested rational lemniscate (86). If 1_P is the canonical rational section of $\mathcal{O}(P)$ and the polar divisor of u_p is $a_p P$, then

$$\epsilon_p = u_p 1_P^{a_p} \tag{146}$$

is nonvanishing on Ω_p° and has capacitary norm identically one there. Indeed, [3, Corollary 5.8] gives $\|1_P\|_p = |u_p|_p^{-1/a_p}$ on $\{|u_p|_p > 1\}$.

Let a source vector have norm at most one and contract it with the horizontal leaf. Write $D = a_p k + b$, $0 \leq b < a_p$. Dividing by ϵ_p^k trivializes the growing power of $\mathcal{O}(P)$; the residual power $\mathcal{O}(bP)$, the coefficient bundle, and the contraction line are fixed. For $r > 1$ in the divisible value group, the maximum-principle computation in the proof of [3, Proposition 5.15], applied on the affinoid $\{|u_p|_p \geq r\}$, gives

$$\log \|\psi^{(n)}_D\|_{p,\mathrm{nat}} \leq C_{p,\mathcal{S}}(r) + \frac{n}{a_p} \log r.$$

By (86), all these affinoids for $r > 1$ sufficiently close to one lie in the fixed affinoid $\{|u_p|_p \geq 1\} \Subset \Omega_p^+$. The horizontal leaf and the fixed contraction pairing are bounded on a neighborhood of that affinoid, while the global source supremum norm bounds the coefficient section there. Hence $C_{p,\mathcal{S}}(r)$ is bounded above by one constant independent of r, D, n. Taking the infimum as $r \downarrow 1$ removes the last term and proves (144). This is the finite-order statement needed here; it is stronger than merely identifying the limiting canonical tangent norm.

For $v \neq p$, the continuation domain is the integral Tate residue disc. By the local model prescription in the construction of $\mathscr{X}_\alpha$, the finite metrics on $P^*\overline{L}_\alpha$ and $\overline{T}_\alpha$ are the same integral target metrics used to define $\delta_{v,D,n}$ in Section 13. Thus (145) is the definition of the positive excess, up to the bounded comparison of the fixed coefficient bundle and contraction line. These comparisons are integral outside a fixed finite set. □

Lemma A.9 (Archimedean Green–Jensen estimate)**.** *Uniformly for every divided jet occurring in the vanishing filtration,*

$$\sum_\sigma \log \|\psi_D^{(n)}\|_{\sigma,\mathrm{nat}} \leq D\mathscr{W}_\alpha + O_{p,\mathcal{S}}(1). \tag{147}$$

Proof. Contract first with the fixed horizontal leaf. On Ω_σ this gives a scalar analytic section s of $\iota_\sigma^* L_\alpha^D$ tensored with a fixed line. Suppose that its first nonzero divided jet at o_σ has order n. Poincaré–Lelong and Green's identity for g_σ give

$$\log \|j_{o_\sigma}^n s\|_{\mathrm{nat}} \leq \int_{\partial\Omega_\sigma} \log \|s\| \, \mathrm{d}\mu_\sigma + D \int_{\Omega_\sigma} g_\sigma \, \iota_\sigma^* c_1(\overline{L}_{\alpha,\sigma}) + O_{p,\mathcal{S}}(1).$$

Every zero away from o_σ occurs with the favorable sign and has been discarded. The source capacitary tangent metric in (142) is exactly the normalization that removes the local derivative term. This is the intrinsic Green–Jensen calculation underlying [7, Lemma 7.4.1].

By Proposition 13.2, the image of the closed pointed domain lies in a slightly larger continuation domain. The horizontal leaf and the fixed coefficient-bundle pairing are therefore bounded on a neighborhood of the boundary. For a source vector of John norm at most one, its supremum norm is at most one because the John ellipsoid is contained in the supremum unit ball. The boundary integral is consequently $O_{p,\mathcal{S}}(1)$, uniformly in D and n. Summing over σ and using the projection formula in (139) proves (147). □

Proposition A.10 (Uniform single-pivot cancellation)**.** *Fix $C > 0$. Uniformly for pivots $0 \leq n \leq CD$,*

$$\widehat{\deg}\overline{J}_{D,n}^\alpha = D\Theta_\alpha - n\Lambda_\alpha + O_{p,\mathcal{S}}(1), \tag{148}$$

$$h_\alpha^{\mathrm{nat}}(\psi_D^{(n)}) \leq D(\mathscr{H}_\alpha - \Theta_\alpha) + \delta_{D,n} + O_{C,p,\mathcal{S}}(1), \tag{149}$$

$$h_\alpha^{\mathrm{sc}}(\psi_D^{(n)}) \leq D\mathscr{H}_\alpha - n\Lambda_\alpha + \delta_{D,n} + O_{C,p,\mathcal{S}}(1), \tag{150}$$

$$\widehat{\deg}\overline{G}_{D,n}^\alpha \leq D\mathscr{H}_\alpha - n\Lambda_\alpha + \delta_{D,n} + O_{C,p,\mathcal{S}}(1). \tag{151}$$

Thus the $D\Theta_\alpha$ target contribution cancels exactly with the $-D\Theta_\alpha$ part of the natural evaluation estimate.

Proof. The target-degree identity follows immediately from (142). Summing Lemma A.8 over the finite places contributes $\delta_{D,n} + O_{p,\mathcal{S}}(1)$. Adding Lemma A.9 and using $\mathscr{W}_\alpha = \mathscr{H}_\alpha - \Theta_\alpha$ proves (149).

Combining (143), (148), and (149) cancels $D\Theta_\alpha$ and proves (150). For the scalarized nonzero map from the rank-one source quotient to the standard degree-zero line, the rank-one degree identity is

$$\widehat{\deg}\overline{G}_{D,n}^\alpha = h_\alpha^{\mathrm{sc}}(\psi_D^{(n)}).$$

This proves (151). □

Remark A.11 (Where overflow is counted). The finite Bost–Chambert-Loir vertical corrections are contained in the target degree $D\Theta_\alpha$. The Archimedean direct-image contribution enters $D\mathscr{W}_\alpha = D(\mathscr{H}_\alpha - \Theta_\alpha)$. Their sum is $D\mathscr{H}_\alpha$. No formal or Archimedean term is inserted a second time. This is the precise target-normal cancellation asserted in Theorem 13.7.

A.5 Hybrid lattices and exact slopes

Proposition A.12 (Exact hybrid assembly). *For the metrics above and the hybrid lattice* (95),

$$\widehat{\deg}\overline{M}_D^{\alpha,\mathrm{hyb}} = \left(\frac{m}{2}\mathscr{H}_\alpha - \xi Q_{\mathcal{S}}\right) D^2 + o_{p,\mathcal{S}}(D^2). \tag{152}$$

Moreover

$$\sum_{n\in V_D} \widehat{\deg}\overline{G}_{D,n}^\alpha \le \left(-\frac{m^2}{2}\Lambda_\alpha + m\mathscr{H}_\alpha\right) D^2 + \mathscr{A}_D + o_{p,\mathcal{S}}(D^2), \tag{153}$$

and joint rationality implies

$$\tau_{p,\mathcal{S}}^{\mathrm{curve}}(\xi) \ge \Lambda_\alpha - \frac{\mathscr{H}_\alpha}{m}. \tag{154}$$

Proof. Use the finite $\mathbf{Q}$-model of $\overline{L}_\alpha$ as the reference source model. At every place $\ell \neq p$, choose the coefficient-bundle models in Section 12 from this same model, after enlarging the fixed exceptional set. Only the fixed metric and integral model of $\mathcal{F}_{\mathcal{S}}$ then require comparison, which changes determinant degrees and summed fixed-place heights by $O_{p,\mathcal{S}}(D\log(D+2))$. At the distinguished place p no comparison with a coefficient lattice is made: it is controlled by the capacitary estimate and is excluded from $\delta_{D,n}$. Passing to a sublattice can only improve its analytic operator norm.

The positive block of weight s has rank $sD + O_{p,s}(1)$. Replacing it by $L_N^s M_{D,s}^+$ therefore changes the degree by

$$-s^2 D\log L_N + O_{p,s}(\log L_N).$$

Since $\log L_N = \xi D + o(D)$, summing over s and using Proposition A.6 proves (152). Shrinking the finite source lattice increases its gauge norm, so the natural jet estimate of Proposition A.10 remains valid.

The residual small-prime heights are $o(D^2)$ by the proper-prime-power argument in Proposition 13.5. At primes $\ell > \xi D$, one has $\ell \neq p$ for $D \gg 1$, and the compatible coefficient models permit the literal application of Propositions 12.3 and 12.4; hence

$$\sum_{n\in V_D} \delta_{D,n} \le s_* m G_{d_p,m}(\xi) D^2 + o_{p,\mathcal{S}}(D^2).$$

By Theorem 11.2, all pivots satisfy $n \le CD$ for one fixed $C = C_{p,\mathcal{S}}$. Sum (151) over the $mD + O(1)$ pivots. Its uniform $O_{C,p,\mathcal{S}}(1)$ remainder contributes only $O_{p,\mathcal{S}}(D)$. Together with (100), this proves (153).

The vanishing filtration has saturated rank-one quotients whose degrees add:

$$\widehat{\deg}\overline{M}_D^{\alpha,\mathrm{hyb}} = \sum_{n\in V_D} \widehat{\deg}\overline{G}_{D,n}^\alpha.$$

Comparing (152) with (153) gives

$$\frac{m}{2}\mathscr{H}_\alpha - \xi Q_{\mathcal{S}} \le m\mathscr{H}_\alpha - \frac{m^2}{2}\Lambda_\alpha + s_* m G_{d_p,m}(\xi).$$

Rearranging and using (98) proves (154). □

Remark A.13 (Logical role). The exact theorem is a canonical refinement of the general-curve slopes argument. The proof of Theorem 1.6 remains based on the robust inequality (99); hence no density conclusion depends on the more delicate exact-normalization appendix.

B Derivation of the capped window

For completeness, let $m \geq 2$ and $1 < \xi < m$. A prime $\ell = xD$ contributes when a pivot lies in a degree-D interval ending at a positive multiple of ℓ. Maximal packing of distinct pivots gives

$$w_m(x) = K_x + (m - (K_x + 1)x)_+, \qquad K_x = \left\lfloor \frac{m-1}{x} \right\rfloor.$$

Integrating w_m gives (5). The function is decreasing and crosses the integer level ρ at $x = m/(\rho+1)$. Therefore

$$\int_\xi^m \min\{w_m(x), \rho\}\, \mathrm{d}x = \rho\left(\frac{m}{\rho+1} - \xi\right) + I^m_{m/(\rho+1)}\left(\frac{m}{\rho+1}\right)$$

when $\xi < m/(\rho+1)$. At the crossing, $K = \rho$ and the final quadratic term vanishes, giving

$$I^m_{m/(\rho+1)} = \left(m - \frac{1}{2}\right) H_\rho - \frac{m\rho}{\rho+1}.$$

This proves (12).

C Certificate methodology

Real intervals are represented by integer endpoints at scale 10^{42} and every operation is rounded outward. The constants are enclosed from

$$\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}, \qquad \log p = 2\,\mathrm{arctanh}\,\frac{p-1}{p+1}.$$

The alternating arctangent remainder is bounded by the first omitted term; the positive arctanh tail is bounded geometrically. For rational $x \in (0,1)$,

$$\arccos x = 2 \arctan \sqrt{\frac{1-x}{1+x}},$$

and square roots are enclosed by exact integer square root. Harmonic sums and all algebraic terms are rational. The active collision layers are selected by the exact test $cY < 1$.

For fixed p, stage k, largest candidate x, and previous-entry sum T, Lemma 9.1 gives the exact maximal Q. The program evaluates the upper endpoint of the arithmetic cost and the lower endpoint of the analytic margin. A strictly positive lower endpoint excludes the entire aggregate state. The stage witnesses and every worst state are printed in the stored output.

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